Chapter 2

Probability

2.1 Introduction

Suppose that an anthropologist wants to classify fossil human skulls into two classes: *male* and *female*. The anthropologist knows from previous experience that male skulls tend to have larger circumferences than female skulls. However, within each group of skulls there is considerable variation in the circumferences and their ranges overlap. If it were necessary to guess the sexes of various skulls given only their circumferences, it would be reasonable to guess *male* whenever the circumference is larger than some value. This is an example of a classification decision (or guess) based on the value of a single feature.

In the field of pattern recognition, we are interested in techniques for making the best decisions possible. Even if we make the best decisions based on the available information, we may not always be correct. If we used additional measurements from the skull, we would probably be correct more often. There are various hereditary, nutritional, and environmental causes for the variations in the circumferences of the skulls, and if we could know them all for each skull, we could conceivably model or explain why these particular circumferences were produced. An even more complex chain of cause and effect would probably be required to explain why these particular skulls were found in our sample. Rather than requiring complete understanding, a **probability model** assumes that at least some of the variability in the data is due to chance or random variability. In this chapter, we will discuss various probability models. They will be used in the following two chapters as the basis for decision making techniques.

For a random occurrence with a finite number of possible outcomes, such as randomly choosing a skull from some group and measuring its circumference x to the nearest centimeter, we can define a probability model by listing all the possible out-

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comes and the probability that each one occurs. We denote the probability that a particular value x occurs by $P(x)$. A random process can be very simple, such as flipping a single unbiased coin. In this case the outcomes are *head* and *tail* and the corresponding probabilities are $P(head) = 1/2$ and $P(tail) = 1/2$. If the experiment consists of rolling a fair die, the outcomes are 1, 2, ..., 6 and the probability of each outcome is 1/6.

Most people have an intuitive notion of the meaning of the word **probability**, but it is used in various ways. A physician might tell a patient that, according to some tests she has made, there is a 10 percent probability that the patient has disease A. Some people would object to the use of the word "probability" in this situation and point out that either the patient really has the disease or he does not, so the unknown probability is either zero or 100 percent and cannot be anything between these two values. They would not apply the word to individual events for which the outcome has already been determined, but only to hypothetical experiments on collections of data or assumed models. According to this point of view, the physician could say that 10 percent of people with these test results had disease A, or that the probability is 10 percent that a randomly chosen past patient with these test results had the disease, but should not say that there is a certain probability that a particular patient has the disease. In this text, whenever we speak of the probability that a sample belongs to a certain class, we do not refer to some particular sample, but to a randomly selected sample that has the same set of feature values as the sample described.

Continuous variables such as the exact circumference of a skull are considered to have an infinity of possible values. Continuous variables can be described by probability densities, which will be discussed in Section 2.3, or their possible values can be broken into ranges, and the probability of lying in each range can be listed. For example, suppose that the probability is 0.02 that the circumference x of a skull is between 39.5 and 40.5 centimeters. We express this as $P(39.5 \leq x \leq 40.5) = 0.02$. A feature from any real data set is automatically broken into a finite number of ranges when it is measured, because measurements can only be recorded with finite precision.

Although choosing a probability model is easy for simple or idealized situations such as flipping unbiased coins or rolling dice with assumed probabilities for various faces showing, obtaining probabilities for real world situations is usually more difficult. Two methods for obtaining probability estimates are called the **frequentist approach** and the **subjective approach**.

In the frequentist approach, the probability of an event is estimated by dividing the number of occurrences of an event by the number of trials. For example, to estimate the probability of obtaining a defective light bulb from a certain manufacturing process, we might sample 100 light bulbs and count the number that are defective. If we obtain three defective light bulbs, we approximate or estimate the probability of obtaining a defective light bulb from the process as $3/100 = 0.03$. We can never know the exact probability; but, in principle, by testing a large enough number of light bulbs, we

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can estimate it as closely and as confidently as desired (short of absolute certainty), assuming that the manufacturing process remains constant.

Although the frequentist approach is easy to understand, it may be difficult or impossible to obtain enough samples to get an estimate of the true probability. Furthermore, the frequentist approach only applies to repeatable events, that is, events for which the probability is constant over all trials. This assumption is often difficult to verify in the real world.

The probability that a certain candidate will win the next presidential election cannot be estimated using the frequentist approach because the event is not repeatable. For events that are not repeatable, a **subjective** assessment may be the only way to assign a probability measure. An intuitively appealing way to quantify the process of selecting a subjective probability is to use the notion of a fair bet. You may not be willing to bet "even money" that it will rain in Las Vegas on the next Fourth of July. In fact, you would prefer to bet against rain because Las Vegas has a desert climate. However, if you were offered 200 to 1 odds, you would probably prefer to bet on rain if you knew that it does rain there occasionally in June or July. Somewhere between the odds of 1 to 1 and the odds of 200 to 1, there is a bet with odds, say 15 to 1, for which you would be equally willing to take either side. Such a bet is called a **fair bet**. Suppose that the probability of the event on which you are betting is P. If the event occurs, you win the amount W, so you expect to win PW on the average per bet (not counting losses). If the event does not occur, you lose L, and since the probability that the event will not occur is $1-P$, you expect to lose an average of $(1-P)L$ per bet (not counting winnings). If the bet is fair, the average amount of money you win will equal the average amount of money you lose, so $PW = (1-P)L$ or $P = L/(L+W)$. Thus the odds of 15 to 1 against rain are equivalent to the probability of $1/(15+1) = 1/16$ that rain will occur and 15/16 that it will not.

We will use mainly the frequentist approach in this book. The subjective approach is, however, useful in pattern recognition, especially when subjective expert opinions are to be incorporated into a decision.

2.2 Probabilities of Events

The term **experiment** is used in probability theory to describe a process for which the outcome is not known with certainty. Examples of experiments are

1. Rolling a fair six-sided die.
2. Randomly choosing ten transistors from a lot of 1,000 new transistors.
3. Selecting a newborn child at St. Luke's Hospital.

An **event** is an outcome or combination of outcomes from a statistical experiment. The theory of probability studies the relative likelihood of the events that might occur

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when an experiment is performed. We will represent events by uppercase letters such as A, B, and C. Examples of events that might occur as a result of performing the previously listed experiments are

1. Obtaining a 6 when a fair six-sided die is rolled.
2. Obtaining an even number when a fair six-sided die is rolled.
3. Finding more than three defective transistors out of ten transistors randomly chosen from a lot of 1,000 new transistors.
4. Selecting a randomly selected newborn child at St. Luke's Hospital weighing more than eight pounds.

The event consisting of all possible outcomes of a statistical experiment is called the **sample space**. We let S denote the sample space. The sample spaces for the previously listed experiments are

1. The sample space consists of the numbers 1, 2, 3, 4, 5, and 6—all possible outcomes of rolling a fair six-sided die.
2. The sample space consists of the numbers $0, 1, \ldots, 10$—all possible numbers of defective transistors that might be obtained when ten transistors are randomly chosen from a lot of 1,000 new transistors.
3. The sample space consists of all numbers that represent the possible weights of randomly selected newborn children at St. Luke's Hospital.

A useful way to visualize relationships among events is to use a **Venn diagram** (see Figure 2.1 for examples). The sample space S is represented by the entire rectangular region.

Events are represented by regions inside the rectangles, and their areas can be made to correspond to the probabilities of events. In Figure 2.1a, the event A is represented by the shaded region. Because exactly one of the outcomes in S must occur in a single trial,

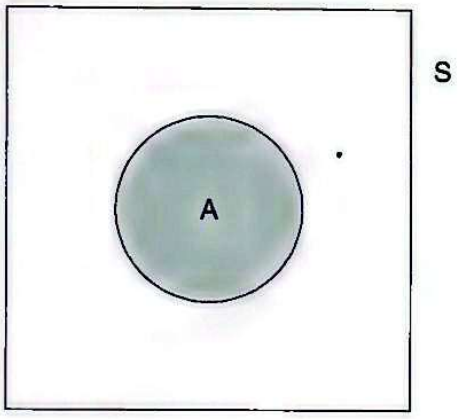
$$P(S) = 1.$$

If A is an event, then the event *not* A is called the **complement** of A and is shown in Figure 2.1b. The complement of A is also denoted by $\overline{A}$. Because it is a certainty that either A or *not* A occurs, $P(A) + P(\textit{not } A) = 1$, so

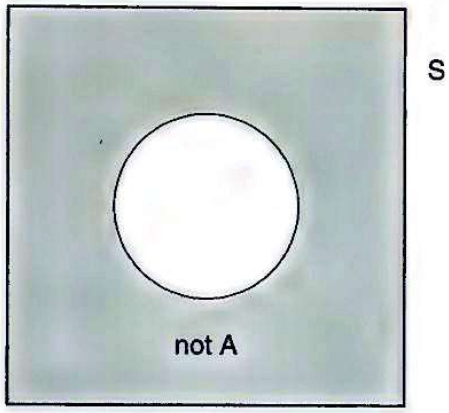
$$P(\textit{not } A) = 1 - P(A).$$

If A and B are events, the event "either A or B or both occurs" is denoted by A *or* B (see Figure 2.1c). This event is sometimes denoted by $A \cup B$ (read "A **union**

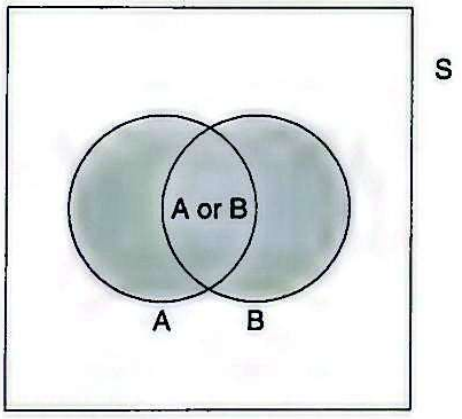
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(a)

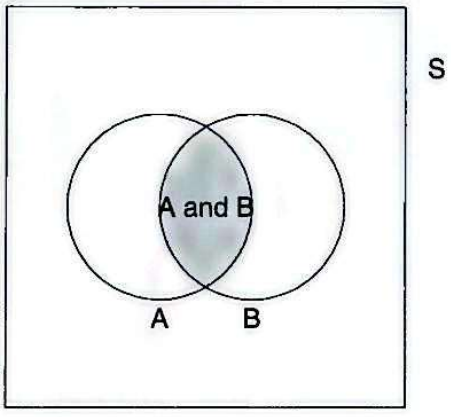


(b)



Union

(c) A OR B
A ∪ B
A + B (real Addition)



Intersection A AND B Joint Event

(d) A ∩ B
AB = A·B (real multiplication)

Figure 2.1: The shaded areas represent the following events: (a) A, (b) $\overline{A}$, (c) A *or* B, (d) A *and* B.

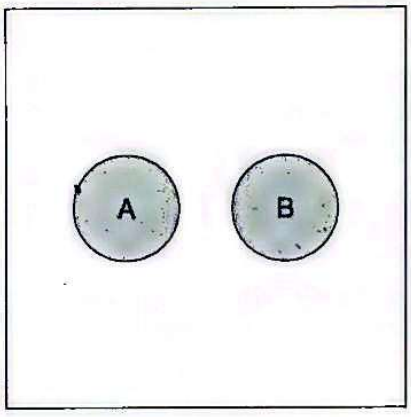


Figure 2.2: Classes A and B are mutually exclusive.

B") or $A + B$. The event "both A and B occur" is denoted by A *and* B (shown in Figure 2.1d). This event is sometimes denoted by $A \cap B$ (read "A **intersection** B") or AB. The event A *and* B is called a **joint event.**

If the event A *and* B cannot occur—that is, A and B cannot occur simultaneously—then the events A and B are said to be **mutually exclusive** (see Figure 2.2). If A and B are mutually exclusive, the following equation called the **addition rule** holds:

$$P(A \text{ or } B) = P(A) + P(B) \tag{2.1}$$

For example, if the probability that x will be the next president is 30 percent and the probability that y will be the next president is 20 percent, then the probability that one of them will be the next president is 50 percent.

If A and B are not mutually exclusive, there are four possible joint events A *and* B, $\overline{A}$ *and* B, A *and* $\overline{B}$, and $\overline{A}$ *and* $\overline{B}$, each of which is mutually exclusive (see Figure 2.3). The addition rule thus applies and we can write

$$\begin{aligned} P(A \text{ or } B) &= P((A \text{ and } B) \text{ or } (\overline{A} \text{ and } B) \text{ or } (A \text{ and } \overline{B})) \\ &= P(A \text{ and } B) + P(\overline{A} \text{ and } B) + P(A \text{ and } \overline{B}) \end{aligned}$$

Furthermore,

$$P(A \text{ and } B) + P(A \text{ and } \overline{B}) = P(A)$$

and

$$P(A \text{ and } B) + P(\overline{A} \text{ and } B) = P(B).$$

These three equations can be combined to yield

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B). \tag{2.2}$$

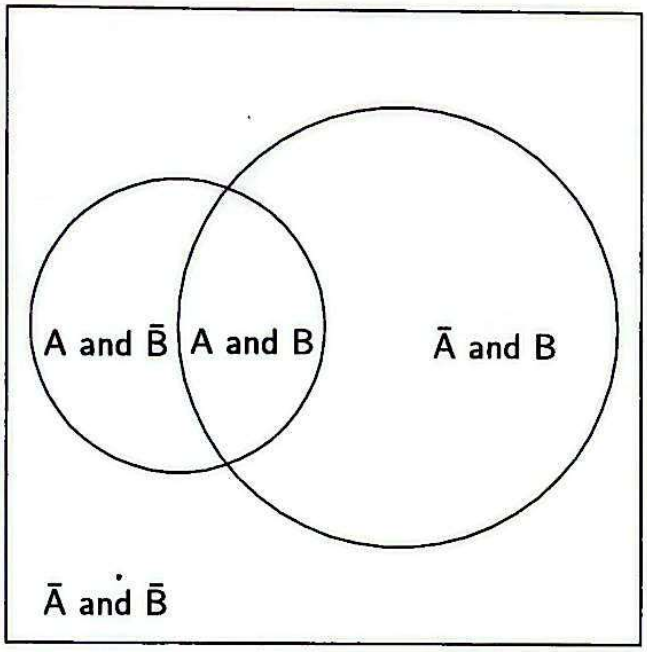


Figure 2.3: A Venn diagram illustrating (2.2).

This is also evident in Figure 2.3. If $P(A)$ is the area of the event A and $P(B)$ is the area of B, then $P(A) + P(B)$ includes the area of the overlapped region $P(A \textit{ and } B)$ twice, so it must be subtracted to obtain the area of the event A *or* B.

Equation (2.2) can be used to compute the probability of drawing an ace or a spade or both from a deck of cards. Because $P(ace) = 1/13$, $P(spade) = 1/4$, and $P(ace \textit{ and } spade) = 1/52$,

$$P(ace \textit{ or } spade) = 1/13 + 1/4 - 1/52 = 4/13$$

Conditional Probabilities

If A and B are events, then the probability of A may depend on whether B occurs. For example, the probability of rolling a 2 with a fair die is 1/6, but if we know that the outcome was an even number, then the probability that a 2 was rolled is 1/3. The **conditional probability** of A occurring, given that B has occurred, is denoted $P(A|B)$ and is read "P of A given B." Since we know in advance that B has occurred, B effectively becomes the new sample space, so $P(A|B)$ is the fraction of the B cases in which A occurs. Thus we obtain the formula

$$P(A|B) = \frac{P(A \textit{ and } B)}{P(B)}. \tag{2.3}$$

This conditional probability is not defined if $P(B) = 0$. Similarly,

$$P(B|A) = \frac{P(B \textit{ and } A)}{P(A)}. \tag{2.4}$$

The expressions (2.3) and (2.4), which can be rewritten as

$$P(A \text{ and } B) = P(B)P(A|B) \tag{2.5}$$

and

$$P(A \text{ and } B) = P(A)P(B|A), \tag{2.6}$$

can also be used to calculate $P(A \text{ and } B)$.

Example 2.1 *Calculating the conditional probability of rain given that the barometric pressure is high.*

Weather records show that high barometric pressure (defined as being over 760 mm of mercury) occurred on 160 of the 200 days in a data set, and it rained on 20 of the 160 days with high barometric pressure. If we let R denote the event "rain occurred" and H the event "high barometric pressure occurred" and use the frequentist approach to define the probabilities, we see that

$$P(H) = 160/200 = 0.80$$

and

$$P(R \text{ and } H) = 20/200 = 0.10.$$

We can obtain the probability of rain, given high pressure, directly from the data:

$$P(R|H) = 20/160 = 0.125.$$

Since it is given that high barometric pressure occurred, only those 160 days of high barometric pressure are under consideration and rain occurred on 20 of those days. We could also obtain $P(R|H)$ from (2.3):

$$P(R|H) = P(R \text{ and } H)/P(H) = 0.10/0.80 = 0.125.$$

Thus (2.3) is consistent with the concept of probability according to the frequentist approach.

Whether we consider these ratios to be estimates or definitions of the various probabilities depends on the population. If we randomly select one of the 200 days in the data set, the probability is exactly 80 percent that there was high barometric pressure on that day. Alternatively, if we consider the population to include all possible days, there is no way to know the exact probability that there will be rain on one of these days. However, we can use past experience to estimate the probabilities of these events on days not included in our data set. In most practical problems, we are interested in estimating the probabilities of future events, given some assumptions or some facts concerning the past or present.

The Multiplication Rule

In many important cases, $P(A)$ may not depend on whether B has occurred. We say that the event A is **independent** of B if $P(A) = P(A|B)$. An important consequence of the definition of independence is the **multiplication rule**, which is obtained by substituting $P(A)$ for $P(A|B)$ in (2.5) to obtain

$$P(A \text{ and } B) = P(A|B)P(B) = P(A)P(B) \tag{2.7}$$

whenever A is independent of B. If A is independent of B, we may combine (2.4) and (2.7) to obtain

$$P(B|A) = \frac{P(B \text{ and } A)}{P(A)} = \frac{P(B)P(A)}{P(A)} = P(B),$$

so if A is independent of B, then B is also independent of A. From now on, we will express either of these conditions as "A and B are independent." If events are not independent, the probabilities of the various possible values of one of them depend on the value of the other.

Suppose we want to calculate the probability of rolling a pair of 1s with two dice. The multiplication rule can be used in this case because the number obtained on one die is independent of the number obtained on the other die. Hence the probability of getting ones on both dice is $(1/6)(1/6) = 1/36$.

Example 2.2 *The addition and multiplication rules.*

The probability of rolling a pair of dice such that the sum of the faces showing will equal three can be found by using the addition and multiplication rules. There are two ways in which the sum of two dice can equal three: Either the first die shows a 2 and the second die a 1, or the first shows a 1 and the second shows a 2. Let F_1 be the event that the first die shows a 1, and let F_2 be the event that the first die shows a 2. Let G_1 and G_2 be the events that the second die shows a 1 or a 2, respectively. Also let E be the event that the sum of the two rolls is 3. If we assume that the rolls of the two dice are independent, the multiplication rule applies. In addition, the events F_1 and F_2 are mutually exclusive, as are G_1 and G_2, so we can use the addition rule:

$$\begin{aligned} P(E) &= P((F_1 \text{ and } G_2) \text{ or } (F_2 \text{ and } G_1)) \\ &= P(F_1 \text{ and } G_2) + P(F_2 \text{ and } G_1) = P(1)P(2) + P(2)P(1) \\ &= (1/6)(1/6) + (1/6)(1/6) = 1/36 + 1/36 = 1/18. \end{aligned}$$

Example 2.3 *Estimating the probabilities of oil recovery test outcomes.*

As an example in which two events are not independent, suppose that a company plans to test two new techniques for improving the extraction of oil from the ground. The first technique consists of setting off an explosion at the bottom of a well to fracture the strata and then testing seismically to determine the extent of the fracturing. The second technique consists of injecting hot brine into the well to loosen the oil and then pumping to measure the oil recovery. Let E be the event that the explosion successfully fractures the strata within a radius of 100 meters, and let R be the event that oil can be recovered at a rate of 50 barrels per day after pumping in hot brine. In a certain region, $P(E)$ is estimated to be 0.8. If E occurs, the probability of R occurring is estimated to be $P(R|E) = 0.9$, but if the explosion is not a success, the estimated probability of recovery is only $P(R|\overline{E}) = 0.3$. These three assumptions are sufficient to define the probabilities of any combination of outcomes, given any set of constraints. The probabilities of the four possible joint events are

$$\begin{aligned} P(E \text{ and } R) &= P(E)P(R|E) = (0.8)(0.9) = 0.72 \\ P(E \text{ and } \overline{R}) &= P(E)P(\overline{R}|E) = (0.8)(0.1) = 0.08 \\ P(\overline{E} \text{ and } R) &= P(\overline{E})P(R|\overline{E}) = (0.2)(0.3) = 0.06 \\ P(\overline{E} \text{ and } \overline{R}) &= P(\overline{E})P(\overline{R}|\overline{E}) = (0.2)(0.7) = 0.14. \end{aligned}$$

Since exactly one of the four joint events E *and* R, $\overline{E}$ *and* R, E *and* $\overline{R}$, or $\overline{E}$ *and* $\overline{R}$ must occur, the sum of the four probabilities above equals 1. These calculations can be arranged in the following table. The numbers 0.8, 0.9, and 0.3 are known, the others are calculated from them.

	$P(E) = 0.8$	$P(\overline{E}) = 0.2$
$P(R) = 0.78$	$(0.9)(0.8) = 0.72$	$(0.3)(0.2) = 0.06$
$P(\overline{R}) = 0.22$	$(0.1)(0.8) = 0.08$	$(0.7)(0.2) = 0.14$

The probability of successful brine recovery was not assumed explicitly, but can be calculated from the assumptions. There are two mutually exclusive ways of having successful brine recovery: either with a successful explosion or without one. Thus,

$$\begin{aligned} P(R) &= P(E)P(R|E) + P(\overline{E})P(R|\overline{E}) \\ &= (0.8)(0.9) + (0.2)(0.3) = 0.72 + 0.06 = 0.78. \end{aligned}$$

Equivalently, $P(R)$ could be obtained by summing the probabilities of the two upper joint events in the table:

$$P(R) = P(E \text{ and } R) + P(\overline{E} \text{ and } R) = 0.72 + 0.06 = 0.78.$$

If only $P(E)$ and $P(R)$ had been known and if we had assumed that they were independent, we would have calculated the probability of both tests succeeding to be

$$P(E \text{ and } R) = P(E)P(R) = (0.8)(0.78) = 0.624.$$

However, from past experience we know that $P(E \text{ and } R) = 0.72$. The multiplication rule does not apply here because the probability of R depends on the outcome of E, so the events E and R are not independent.

Some other questions are

1. What is the probability that the explosion and the brine tests are both successful or both fail? This equals

$$\begin{aligned} P((E \text{ and } R) \text{ or } (\overline{E} \text{ and } \overline{R})) &= P(E \text{ and } R) + P(\overline{E} \text{ and } \overline{R}) \\ &= 0.72 + 0.14 = 0.86. \end{aligned}$$

2. What is the probability that the explosion test was successful, given the constraint that only one of the two tests was successful? The probabilities of the two events that have exactly one success are $P(E \text{ and } \overline{R}) = 0.08$ and $P(\overline{E} \text{ and } R) = 0.06$, so

$$P(\text{only one success}) = 0.08 + 0.06 = 0.14,$$

and

$$P(E \text{ and only one success}) = P(E \text{ and } \overline{R}) = 0.08.$$

Thus

$$\begin{aligned} P(E|\text{only one success}) &= P(E \text{ and only one success})/P(\text{only one success}) \\ &= 0.08/0.14 = 0.571. \end{aligned}$$

3. What is the probability that the explosion was successful, given that recovery using brine was successful? This is

$$\begin{aligned} P(E|R) &= P(E \text{ and } R)/[P(E \text{ and } R) + P(\overline{E} \text{ and } R)] \\ &= 0.72/(0.72 + 0.06) = 0.923. \end{aligned}$$

4. If one or more of the tests was successful, what is the probability that the other one was successful? The answer is

$$P(E \text{ and } R)/[P(E \text{ and } R) + P(\overline{E} \text{ and } R) + P(E \text{ and } \overline{R})]$$

$$= 0.72/(0.72 + 0.08 + 0.06) = 0.837.$$

2.3 Random Variables

An important first step in pattern recognition is making measurements or extracting features to use for classifying the patterns. For example, one might measure the circumference of a fossil skull that one is trying to classify or count the number of teeth. A **random variable** is the outcome of a random process which outputs a numeric value. The output of a random variable is called a **random number**. An example of a random variable is the process of randomly choosing a sample from some population and measuring one of its features. We will denote the names and values of random variables by lower-case letters such as x, y, and z.

An important special class of random variables are the **discrete random variables** which can take on a finite number of possible values or a countably infinite number of values such as 0, 1, 2, ... or 2, 4, 6, ... A discrete random variable is described by its **distribution function** which lists for each outcome x the probability $P(x)$ of x. If $x_1, x_2, \ldots, x_n$ are all the possible outcomes, then

$$\sum_{i=1}^{n} P(x_i) = 1 \tag{2.8}$$

because some outcome must occur.

Example 2.4 *The distribution function for the number of heads from two flips of a coin.*

The random variable k is defined to be the total number of heads that occur when a fair coin is flipped two times. This random variable k can have only the three possible values 0, 1, and 2, so it is discrete. Its distribution function can be obtained by noting that there are four equally likely outcomes from the two flips: (T,T), (H,T), (T,H), (H,H). One of these outcomes produces no heads, two produce one head, and one produces two heads. The probabilities $P(k)$ of the possible values of k are thus

k	$P(k)$
0	1/4
1	1/2
2	1/4

$= 2/4$

The Binomial Distribution

A fixed length sequence of events or trials where each event has exactly two possible outcomes can be modeled by a **binomial distribution**. One of the outcomes is generally called **success** and the other is called **failure**. In the binomial distribution, we assume that the probability of success is the same for each trial and we denote this probability by θ. The total number of successes k obtained in n trials is called a **binomial random variable**. Its distribution function is given by

$$P(k) = \binom{n}{k}\theta^k(1-\theta)^{n-k}, \quad k = 0, \ldots, n \tag{2.9}$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The symbol $\binom{n}{k}$ is called the **binomial coefficient**.

To see how the binomial distribution is obtained, notice that the n trials will generate a sequence of successes and failures of length n. We want to find the probability of obtaining a sequence that contains k successes and $n-k$ failures. There may be many different sequences that contain k successes and $n-k$ failures. For any particular one of these sequences, we can use the multiplication rule to find the probability of the sequence, as all the trials are independent. From the multiplication rule, we see that the probability of obtaining any particular sequence containing k successes and $n-k$ failures is

$$\theta^k(1-\theta)^{n-k}.$$

Furthermore since the number of different sequences that contain k successes and $n-k$ failures is the number of ways of choosing a set of k samples from a set of n, or $\binom{n}{k}$, the probability that a random sequence will contain k successes and $n-k$ failures can be found from the distribution function (2.9). The number of heads k in Example 2.4 is a binomial random variable because each flip has only two possible outcomes. Its parameters are $n = 2$ and $\theta = 0.5$. In Example 2.5, k is a binomial random variable, where $n = 6$ and $\theta = 0.6$. Its mean μ_k is $n\theta$ and its variance (see Section 2.5) σ_k^2 is $n\theta(1-\theta)$.

Example 2.5 *The binomial distribution.*

An anthropologist knows from records kept at a dig site that the probability that a recovered fossil human skull is female is 0.6. The probability that out of six skulls, exactly four will be female is given by substituting into the binomial distribution function

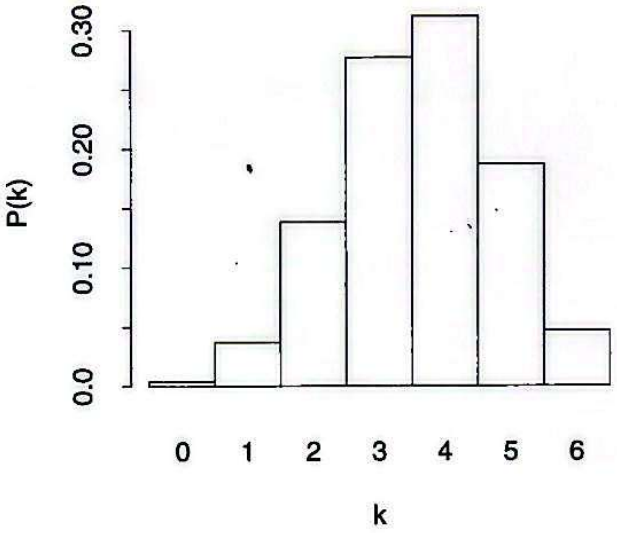


Figure 2.4: The binomial distribution function $P(k)$ for $n = 6$ and $\theta = 0.6$.

$$P(4) = \binom{6}{4}(0.6)^4(0.4)^2 = 0.311.$$

The graph of the binomial distribution function $P(k)$ when $n = 6$ and $\theta = 0.6$ is shown in Figure 2.4.

The Poisson Distribution

Although the discrete random variables we have discussed so far were defined to have only a finite number of possible values, there are random variables that may have a countably infinite number of outcomes. A set of outcomes is countably infinite if the outcomes can be listed in an infinite sequence. For example, the set of nonnegative integers is countably infinite because they can be listed as the infinite sequence $\{0, 1, 2, \ldots\}$. The **Poisson distribution** (see Figure 2.5)

$$P(n) = \frac{e^{-\lambda}\lambda^n}{n!} \tag{2.10}$$

where $n = 0, 1, 2, \ldots$ is an example of such a discrete distribution. The value $P(n)$ is interpreted as the probability that exactly n events will occur in a fixed time interval and λ is the number of events that occur in that length of time on the average, so λ is the mean of the random variable n. We assume that the events occur randomly and independently with a constant probability of occurring in any small time interval, and

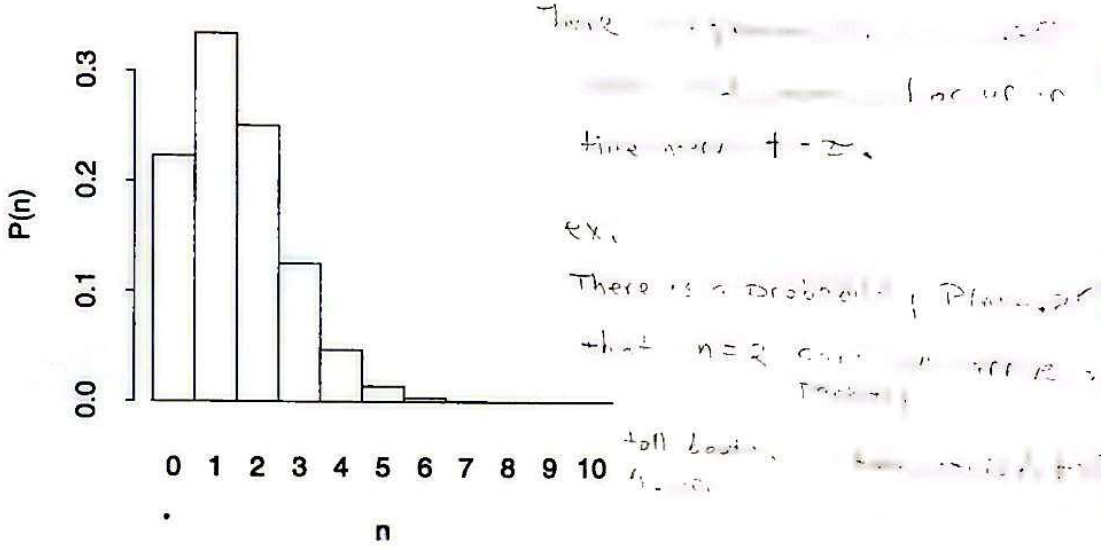


Figure 2.5: The Poisson distribution: $P(n) = e^{-\lambda}\lambda^n/n!$, for $\lambda = 1.5$. $P(n) > 0$ for all n, but the values of $P(n)$ for $n > 6$ are too small to see in the graph.

that two events cannot occur at exactly the same time. The Poisson distribution can be used to model the number of arrivals of input or output requests at a telephone exchange, of automobiles arriving at a tollbooth, of radioactive particles entering a Geiger counter, and of many other processes occurring in scientific and engineering applications.

Like the binomial distribution, the Poisson distribution assumes that only integer values are possible, but unlike the binomial distribution, there is no upper bound on the number of events that can occur in a given time interval for the Poisson distribution. There is a finite but very small probability that any extremely large value of n will occur. It can be shown that both the mean and variance of the Poisson distribution equal the parameter λ (see Section 2.5).

Continuous Random Variables

Discrete random variables can take on only a finite number of values or at most a countable infinity of values (such as the Poisson distribution). However, many random variables observed in everyday life are not discrete but can take on any value in some finite or infinite range. For example, the highest temperature on a given day, the length of a part being inspected, the average intensity of light falling on a given region of an object, or the waiting time for a radioactive count are all random variables that are continuous rather than discrete.

A **continuous random variable** is described by a **probability density function**. This function is used to obtain the probability that the value of a continuous random variable is in a given interval. If the random variable is x and its density

function is $p(x)$, then by definition

$$P(a \leq x \leq b) = \int_a^b p(x)\,dx. \tag{2.11}$$

An example is shown in Figure 2.6. The shaded area equals the probability that x lies between a and b. Density functions are sometimes also called distributions.

While distributions of discrete random variables must sum to 1 (see (2.8)), densities of continuous random variables must integrate to 1, because the probability is 1 that x lies between $-\infty$ and ∞:

$$\int_{-\infty}^{\infty} p(x)\,dx = 1. \tag{2.12}$$

In this text, we will always use uppercase P to represent probabilities such as $P(k)$, and lowercase p to represent densities such as $p(x)$.

A random variable x can also be defined by its **cumulative distribution function** C which is given by

$$C(a) = P(x \leq a).$$

Thus $C(a)$ is the probability that x is less than or equal to a. Using (2.11),

$$C(a) = \int_{-\infty}^{a} p(x)\,dx. \tag{2.13}$$

For any cumulative distribution function, $C(-\infty) = 0$ and $C(\infty) = 1$. Substituting (2.13) into (2.11) gives

$$P(a \leq x \leq b) = \int_a^b p(x)\,dx = \int_{-\infty}^b p(x)\,dx - \int_{-\infty}^a p(x)\,dx = C(b) - C(a). \tag{2.14}$$

Even if $p(x)$ cannot be integrated analytically, $P(a \leq x \leq b)$ can still be obtained numerically using (2.14) if a table of $C(x)$ is available.

Equation (2.11) shows that the probability of occurrence of a particular value of a continuous random variable x, say a, is

$$P(x = a) = P(a \leq x \leq a) = \int_a^a p(x)\,dx = 0.$$

Thus the probability of outcome a is 0, even though the outcome $x = a$ is technically possible. However, when using a continuous random variable x such as the highest temperature on a given day, one is not interested in knowing the probability that x attains some exact value with infinite precision such as $P(x = 85.0035741\ldots)$ but rather that x falls in a finite range such as $P(84.01 \leq x < 84.02)$. Such probabilities are computed by (2.11).

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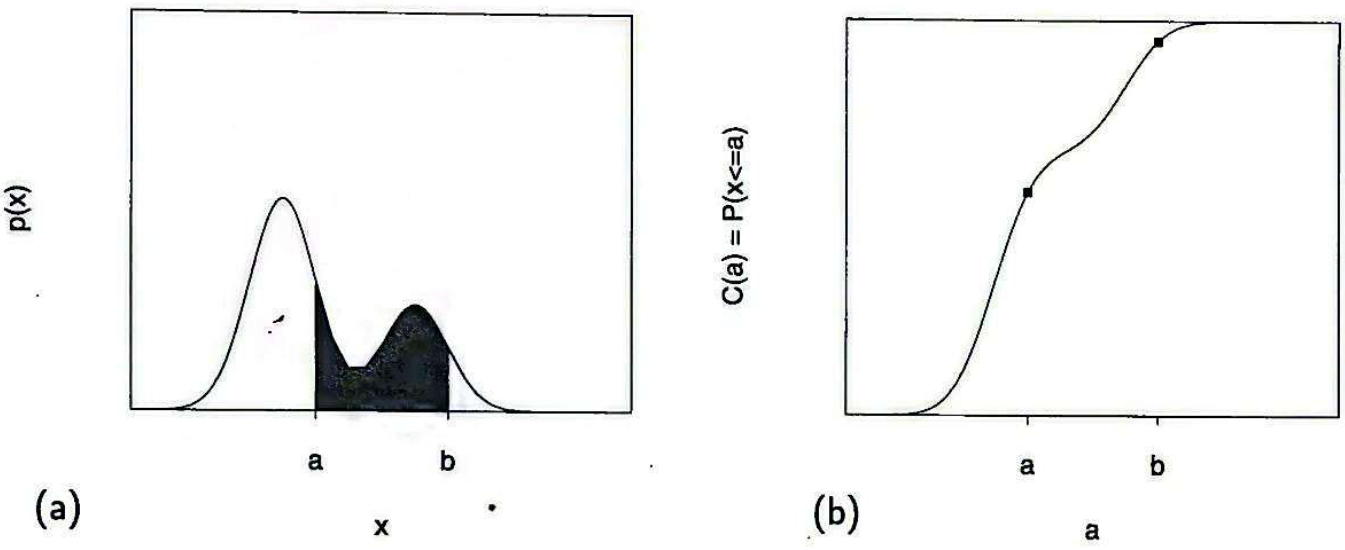


Figure 2.6: (a) A density function. The shaded area is the probability that x lies between a and b. (b) The cumulative distribution function for (a).

If the cumulative distribution function is known and can be differentiated everywhere, the probability density function can be obtained by

$$p(x) = \frac{d}{da}C(a)\Big|_{a=x} = \frac{d}{da}\int_{-\infty}^{a} p(x)\,dx.$$

This formula is a consequence of the fundamental theorem of calculus which asserts that differentiation and integration are inverse operations.

The cumulative distribution function is also defined for a sample of real data. To produce Figure 2.7, 25 16-digit numbers in the range 0 to 1 were generated by a computer using a random number generator. They were then sorted from smallest to largest. For any value of x less than the smallest value, the cumulative distribution of the sample is 0. For x lying between the smallest and next smallest, $C(x) = 1/25$. At each sample value, the cumulative distribution function increases by $1/25$. Most of the cumulative distribution functions for the continuous random variables discussed in this chapter can be differentiated to yield densities, but the distributions for finite data sets, such as the function in Figure 2.7, are discontinuous and cannot be differentiated to yield density functions. In fact, after the data points have been obtained, they are not described by a density but by a discrete distribution, to which each sample contributes a probability of $1/25$.

The Uniform Density

A **uniform** or **rectangular** random variable x is equally likely to attain any value in an interval $[a, b]$. (This interval is also called the **range** of x.) The density of x is

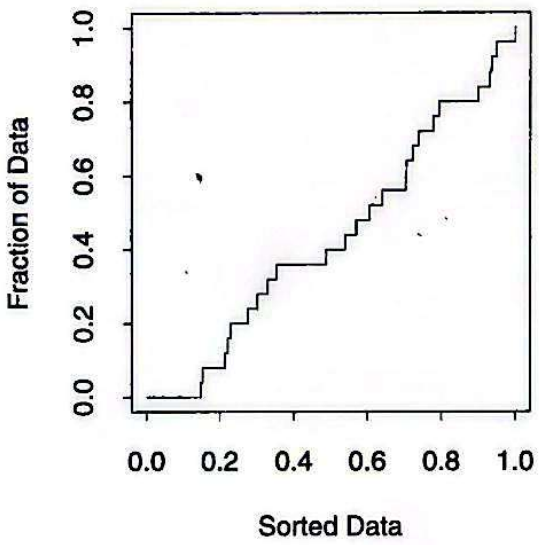


Figure 2.7: Cumulative distribution function for a sample of 25 random numbers chosen from a uniform density.

given by

$$p(x) = \begin{cases} 0 & \text{if } x < a \\ 1/(b-a) & \text{if } a \le x \le b \\ 0 & \text{if } x > b. \end{cases} \tag{2.15}$$

The denominator $b - a$ makes the total area under the density 1. Integrating the density in (2.15) gives the cumulative probability distribution

$$C(x) = \begin{cases} 0 & \text{if } x < a \\ (x-a)/(b-a) & \text{if } a \le x \le b \\ 1 & \text{if } x > b. \end{cases} \tag{2.16}$$

The graphs of (2.15) and (2.16) for $a = 0$ and $b = 1$ are shown in Figure 2.8.

Example 2.6 *The uniform density.*

To obtain the probability that x is between 0.2 and 0.7, if x is uniformly distributed in the interval $[0, 1]$, we can use (2.15):

$$P(0.2 \le x \le 0.7) = \int_{0.2}^{0.7} \frac{1}{1-0}\, dx = 0.5,$$

or we can use (2.16):

$$P(0.2 \le x \le 0.7) = C(0.7) - C(0.2) = 0.7 - 0.2 = 0.5.$$

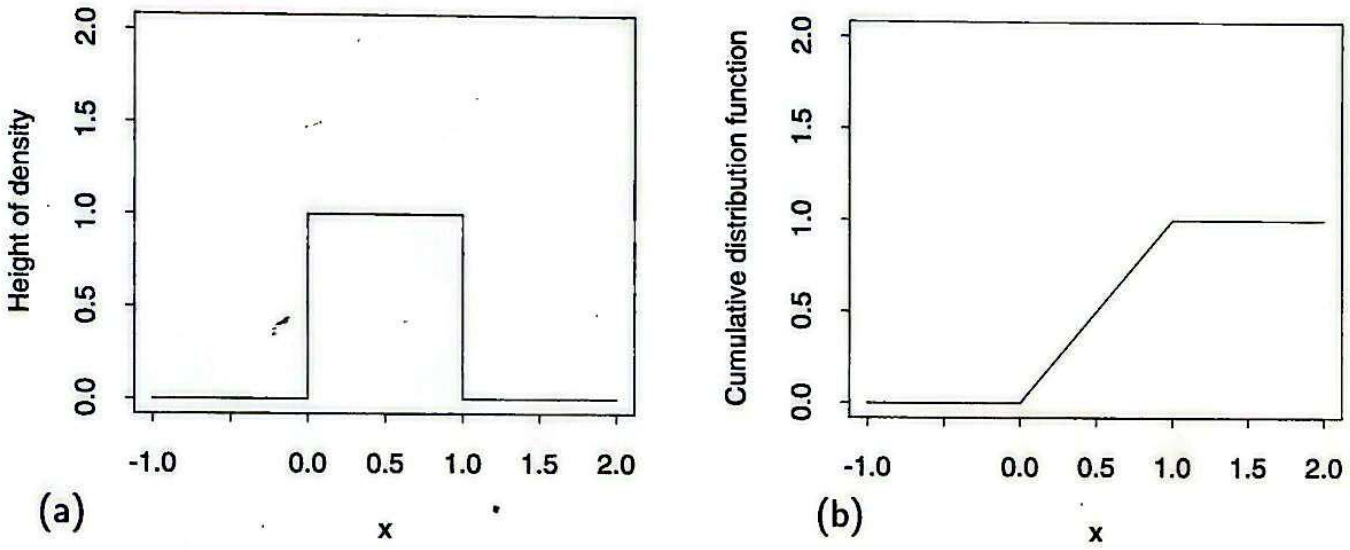


Figure 2.8: The uniform distribution for $a = 0$ and $b = 1$. (a) Density function. (b) Cumulative distribution function.

The Exponential Density

A continuous density that is important in areas such as the study of the reliability of systems and the decay of radioactive particles is the **exponential density** which has the form

$$p(t) = \beta e^{-\beta t}, \quad t \geq 0$$

where t is the time at which an event occurs, such as the decay of a radioactive atom; it could be any value from 0 to ∞. The parameter β is the rate constant, which controls the rate of decay. The cumulative distribution function

$$C(t) = \int_0^t \beta e^{-\beta t}\, dt = 1 - e^{-\beta t}$$

is the probability (at time 0) that the event will occur before time t. The graphs of the density and cumulative distribution for the exponential distribution are shown in Figure 2.9.

Example 2.7 *The exponential density.*

Suppose that the lifetime of a particular type of radioactive atom in years has an exponential density with $\beta = 0.01$. The probability that the atom will decay within 50 years after year 0 is

$$P\{0 \leq t \leq 50\} = \int_0^{50} 0.01 e^{-0.01t}\, dt = 1 - e^{-0.5} = 0.39.$$

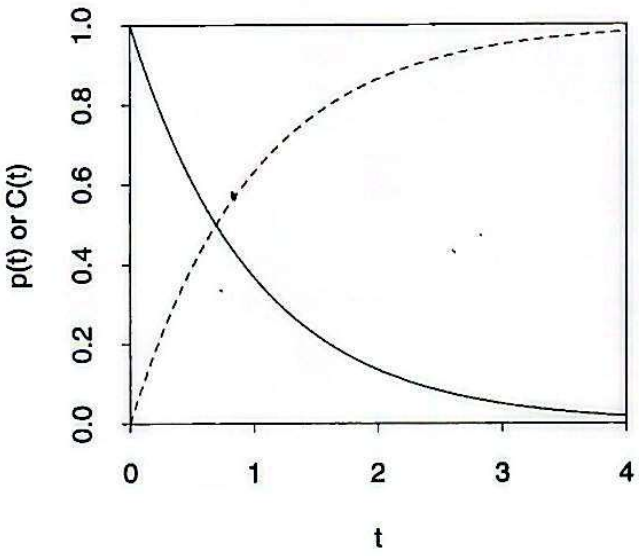


Figure 2.9: The exponential distribution when $\beta = 1$. The density function $p(t) = \beta e^{-\beta t}$ is the solid curve, and the cumulative distribution function $C(t) = 1 - e^{-\beta t}$ is dashed.

In general, the probability that a given atom still remains at time t is $e^{-\beta t}$, so the probability that it decays before t is $1 - e^{-\beta t}$.

Instead of expressing the lifetime distribution of a radioactive element in terms of the parameter β, it is traditional to express it by its **half-life** λ_0, which is defined as the time for half of a given sample to decay, or

$$P(0 \leq t \leq \lambda_0) = 1/2.$$

Since

$$P(0 \leq t \leq \lambda_0) = \int_0^{\lambda_0} \beta e^{-\beta t}\, dt = 1 - e^{-\beta \lambda_0} = 1/2,$$

we obtain $\lambda_0 = (\ln 2)/\beta$ or $\beta = (\ln 2)/\lambda_0$. In this example, the half-life of the atomic isotope to which the atom belongs is $\lambda_0 = (\ln 2)/0.01 = 69.31$ years.

The Normal Density

A very important density is the **normal density** (sometimes called the **Gaussian density**). Its equation is

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{2.17}$$

as shown in Figure 2.10. The parameter μ denotes the **mean** of x, while σ denotes the **standard deviation** of x. Both the mean and the standard deviation are defined in

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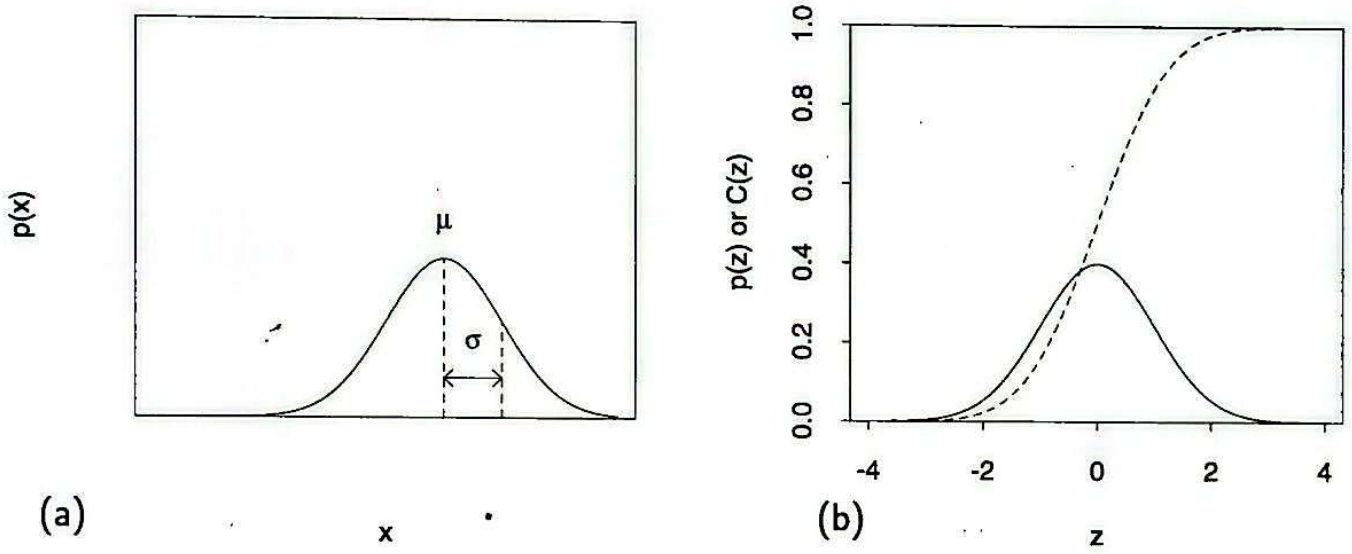


Figure 2.10: (a) A normal density with mean μ and standard deviation σ. (b) The standard normal density function $p(z) = e^{-z^2/2}/\sqrt{2\pi}$ (solid curve), and its cumulative distribution function (dashed curve).

Section 2.5. For a normal distribution, μ is located at the center of symmetry of the normal density, and it can be shown that σ is the distance from the mean to one of the inflection points of the density.

The normal cumulative distribution function is defined by

$$C(s) = \int_{-\infty}^{s} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx,$$

which cannot be integrated in closed form. However, its values are available in mathematical tables and on some scientific calculators, where it is often called the "cumulative error function." Fortunately, a separate table is not required for every value of μ and σ. The special case when $\mu = 0$ and $\sigma = 1$ is called the **standard normal density**:

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \tag{2.18}$$

Its graph is shown in Figure 2.10b. Any normally distributed random variable x can be transformed to a random variable z which has the standard normal density by using the substitution $z = \frac{x-\mu}{\sigma}$. This reduces $P(a \le x \le b)$ to $P\left(\frac{x-a}{\sigma} \le z \le \frac{x-b}{\sigma}\right)$, which is

$$\begin{aligned} \int_a^b p(x)\,dx &= \int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \int_{(a-\mu)/\sigma}^{(b-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz \\ &= C\left(\frac{b-\mu}{\sigma}\right) - C\left(\frac{a-\mu}{\sigma}\right). \end{aligned}$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.998650									
3.5	0.9998674									
4.0	0.99996833									
4.5	0.999996602									
5.0	0.9999997133									
5.5	0.99999998101									
6.0	0.999999999013									
6.5	0.9999999999588									
7.0	0.99999999999872									

$$C(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz$$

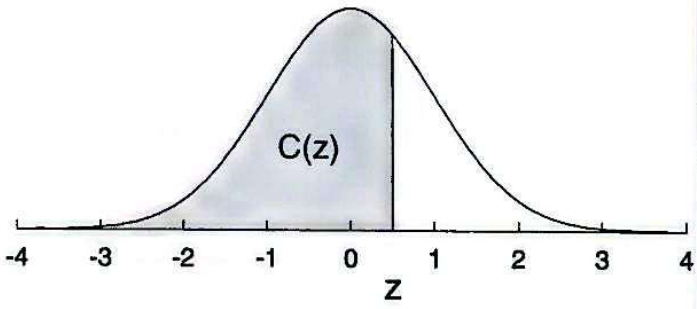


Figure 2.11: Areas under the standard normal curve from $-\infty$ to z, where $z \geq 0$.

z	0.00	−0.01	−0.02	−0.03	−0.04	−0.05	−0.06	−0.07	−0.08	−0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−3.0	1.350×10^{-3}									
−3.5	2.326×10^{-4}									
−4.0	3.167×10^{-5}									
−4.5	3.398×10^{-6}									
−5.0	2.867×10^{-7}									
−5.5	1.800×10^{-8}									
−6.0	9.866×10^{-10}									
−6.5	4.016×10^{-11}									
−7.0	1.280×10^{-12}									

$$C(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz$$

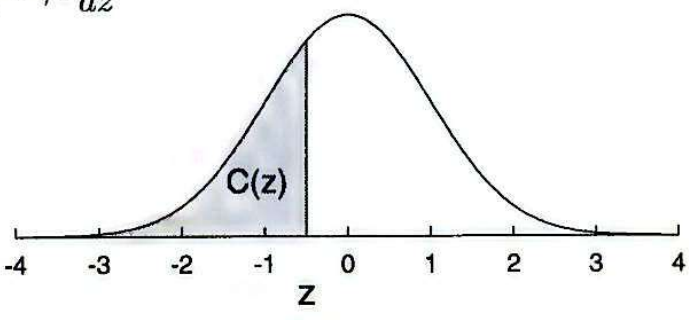


Figure 2.12: Areas under the standard normal curve from $-\infty$ to z, where $z \leq 0$. For $z < -5$, $C(z) \approx (z^2 - 1)e^{-z^2/2}/(-z^3\sqrt{2\pi})$ with an error of less than 1 percent.

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Tables of $C(z)$ for $z \geq 0$ and $z \leq 0$ are given in Figures 2.11 and 2.12. (Actually only one of these tables is needed; the normal density is symmetric and has unit area, so $C(z) = 1 - C(-z)$ for $z < 0$.) The graph of $C(z)$ is shown as a dashed curve in Figure 2.10b.

The transformation that standardizes the normal density

$$z = \frac{x - \mu}{\sigma} \tag{2.19}$$

is known as the **standardizing transformation.**

Example 2.8 *The normal density.*

Consider an electrical circuit in which the voltage is normally distributed with mean 120 and standard deviation 3. What is the probability that the next reading will be between 119 and 121 volts?

The z-transformation is

$$z = (x - 120)/3,$$

so we obtain from Figure 2.11 or 2.12

$$\begin{aligned} P(119 \leq x \leq 121) &= C\left(\frac{121 - 120}{3}\right) - C\left(\frac{119 - 120}{3}\right) \\ &= C(0.333) - C(-0.333) \\ &= 0.631 - (1 - 0.631) = 0.631 - 0.369 = 0.262. \end{aligned}$$

The value 0.631 was obtained by linear interpolation between $C(0.33)$ and $C(0.34)$ in Figure 2.11.

Choosing a Probability Distribution

In some situations, the nature of the data source determines what type of density function is appropriate to the problem. In other cases, theory offers no guide and simply looking at a histogram of the data may suggest a model. Any assumed distribution can be tested using the chi-squared test or the Kolmogorov–Smirnov test to see how likely it would be to produce data that deviate from their expected values as much as do the data actually obtained. Further discussion of these topics is available in many statistics texts.

Although by far the most frequently used continuous density is the normal density, there is usually no particular reason to believe that a given feature will be even approximately normally distributed. The real world never fits any simple model exactly.

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Moreover, the normal model assumes that the feature value can range from $-\infty$ to $+\infty$ which is usually unrealistic. If a feature is normally distributed, any linear function of that feature will also be normally distributed, but nonlinear functions of it will not be. For example, if particle "size" is considered to be important, the most pertinent features could be the diameters, the surface areas, or the volumes of the particles, depending on the application. If the surface areas of a collection of spheres are normally distributed, then neither their diameters nor their volumes can be normally distributed because they are both nonlinear functions of the area. The area is proportional to the square of the diameter and the volume is proportional to its cube. Thus the normality of the data would depend on which of these measures we used, and we should not expect most features to be normal.

As this example suggests, a feature that is not normally distributed can sometimes be made approximately normal by a suitable nonlinear transformation. However, one must be careful not to overfit the data by using a transformation that is too complicated. This transformation may cause the next transformed data set to depart even farther from a normal distribution. In addition, making a multivariate data set normal in each feature still does not guarantee that the set of features taken together will be multivariate normal.

In some cases where a normal distribution does not fit well, densities that have **thicker tails** may be useful. A density $p(x)$ is said to have **thicker tails** than a density $q(x)$ if $p(x) > q(x)$ whenever x approaches ∞ or $-\infty$. An extreme example of such a density is the **Cauchy density** given by

$$p(x) = \frac{1}{\pi b \left(1 + \frac{x-a}{b}\right)^2}. \tag{2.20}$$

This density has been used to model the scattering of electrons as they pass through thin metal foil. Another example is the **double exponential density** given by

$$p(x) = \frac{1}{2b} e^{-|(x-a)/b|}.$$

The normal, Cauchy, and double exponential distributions are compared in Figure 2.13a, where they have been normalized to contain the same area (0.68) between -1 and 1. These densities are discussed in the problems at the end of this chapter. The corresponding cumulative distributions are compared in Figure 2.13b.

A popular graphical test for determining whether a data set is approximately normally distributed is based on a **normal plot**. This plot is formed by plotting the sorted sample values versus the **expected z-values**, that are the values of z which divide the area under the standard normal curve into $n+1$ equal areas, where n is the number of samples. If there are only five samples, we might expect them to lie near the five boundaries between six equal areas on the average, if they actually are drawn from a normal distribution. (See Section 2.6 for further discussion.)

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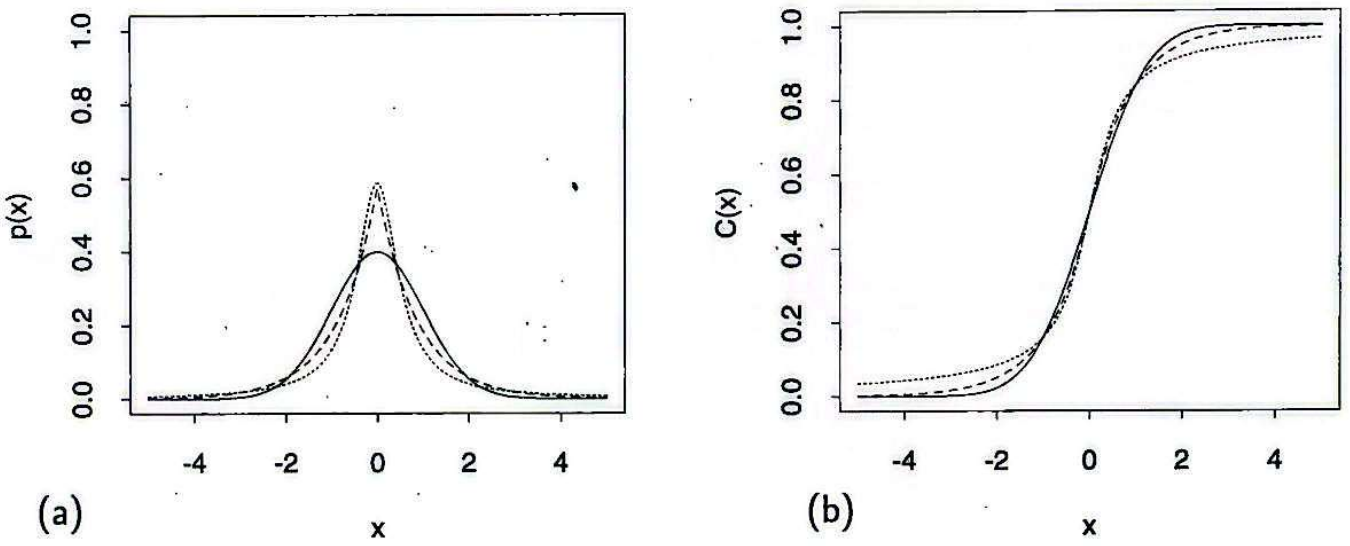


Figure 2.13: A comparison of the normal (solid), double exponential (dashed), and Cauchy (dotted) densities. (a) Densities and (b) cumulative distribution functions.

For example, to construct the normal plot for the samples 3, 1, 2, 9, and 5, sort the samples to get 1, 2, 3, 5, and 9 and plot them versus the points -0.967, -0.430, 0.000, 0.430, and 0.967 which divide the standard normal curve in Figure 2.14a into 5+1 equal areas of 1/6 each. For example, $C(-0.967) = 1/6$ and $C(-0.430) = 2/6$, as found in the tables of Figures 2.11 and 2.12. This produces the normal plot in Figure 2.14b. If a large data set is normally distributed, we can expect that 1/6 of the data will have $z < -0.967$, 2/6 will have $z < -0.430$, 3/6 will have $z < 0$, 4/6 will have $z < 0.430$, and 5/6 will have $z < 0.967$. For a small data set these z values will only be approximate.

If the data are approximately normally distributed, the points on the normal plot will be close to a straight line as in Figure 2.15a. The slope and intercept of the line will depend on the mean and standard deviation of the distribution of the data. The points in the normal plot in Figures 2.15b and 2.15c are not well fit by a line, but the data in Figure 2.15a, which came from a normal density, are well fit by a line. If the samples are drawn from a Cauchy distribution, which has thicker tails than a normal distribution, the normal plot shown in Figure 2.15b turns down at the left end and up at the right end. The opposite is true in Figure 2.15c, where the samples are taken from a uniform density, which has thinner tails than a normal density.

Note that the normal plot can be used to compare data with any mean and variance to a normal distribution, even though the mean and variance of the standard normal distribution are 0 and 1 respectively, because the ability to fit the points in the normal plot with a straight line is independent of the scale of the vertical axis. Although the vertical axis of the normal plot will be translated or stretched, depending upon the

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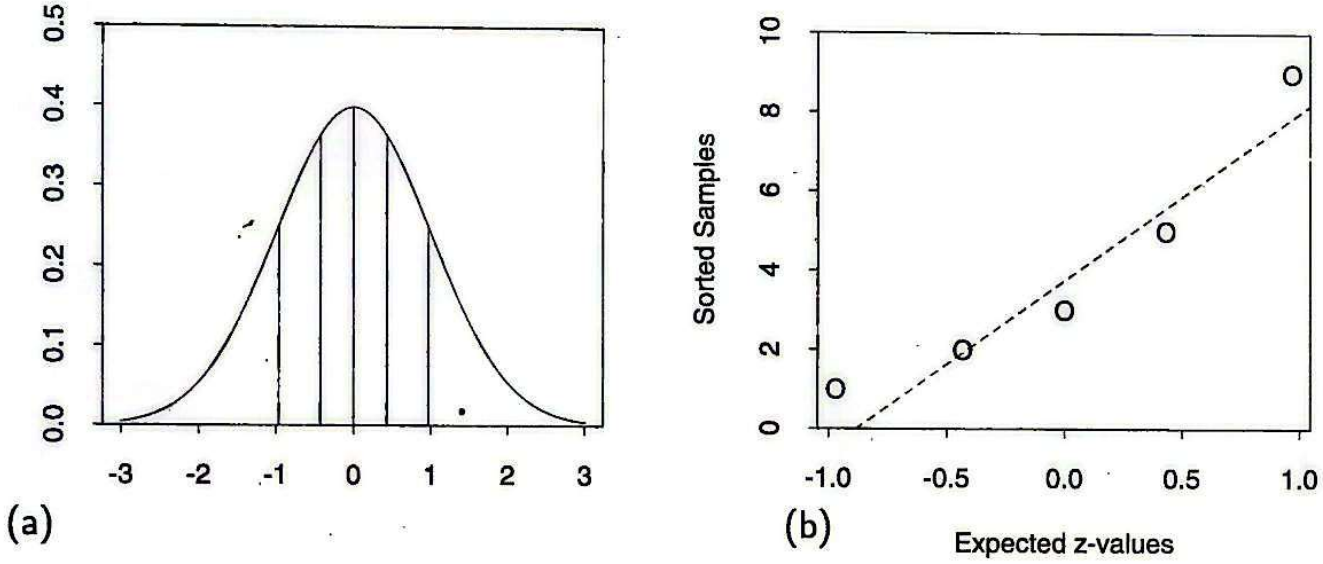


Figure 2.14: (a) A standard normal density divided into six equal areas. (b) A normal plot of five numbers and a linear approximation to them.

mean and variance of the samples, the normal plot will approximate a straight line if the data are close to normal. If s represents the ranked sample values, the mean value of s will be at $z = 0$ on the fitted line. In Figure 2.14b, a line has been fit to the data (by eye). At $z = 0$, s is about 4, so if the data really did come from a normal distribution, it probably has a mean near 4.

The normal plot can be generalized to compare the sample data to *any* distribution, not necessarily normal. This is done by choosing the points on the horizontal axis (called **quantiles**) which divide the density function of that particular distribution into $n + 1$ equal areas, and plotting the ranked values of the samples on the vertical axis as before. Such a plot is called a **quantile-quantile plot** of which the normal plot is a special case. Other tests of the goodness of fit of a theoretical distribution to a data set are described in statistics texts.

2.4 Joint Distributions and Densities

Although it is useful to model the variation of a single feature within a class by a random variable, it is frequently necessary to model the simultaneous variation of two or more features. The joint random variable (x, y) signifies that, simultaneously, the first feature has the value x and the second feature has the value y. If the random variables x and y are discrete, the **joint distribution function** of the joint random

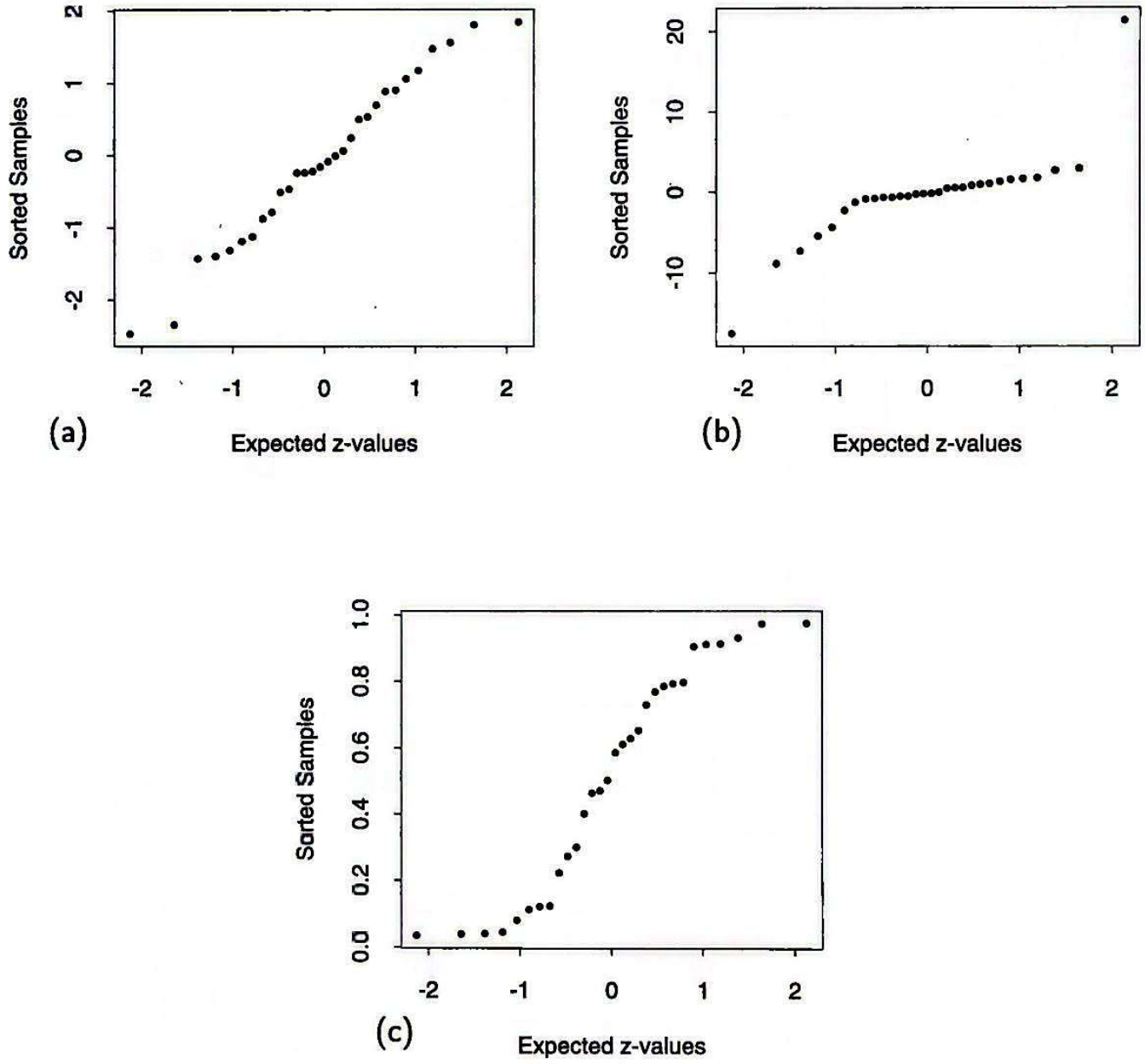


Figure 2.15: A normal plot of 30 numbers drawn from three distributions. (a) Normal, (b) Cauchy, (c) uniform.

variable (x, y) is the probability $P(x, y)$ that both x and y occur. Thus, the joint distribution gives the probability of every possible combination of outcomes of the random variables that make up the joint random variable.

Example 2.9 *The joint distribution function of the outcomes of flipping biased coins.*

A biased coin A has $P(head) = 0.6$ and coin B has $P(head) = 0.3$. Suppose that we flip coin A, and if the outcome is *head*, we flip coin B, but if the result of the first flip is *tail*, we flip coin A again. Since the probability of *head* on the second flip depends on the outcome of the first flip, the two events are not independent. Let x be the outcome of the first flip and let y be the outcome of the second flip. The joint distribution $P(x, y)$ is

(x, y)	$P(x, y)$
$(head, head)$	$(0.6)(0.3) = 0.18$
$(head, tail)$	$(0.6)(0.7) = 0.42$
$(tail, head)$	$(0.4)(0.6) = 0.24$
$(tail, tail)$	$(0.4)(0.4) = 0.16$

A continuous one-dimensional random variable can take on any value in an interval, but a continuous two-dimensional random variable can take on any value in a two-dimensional region. As in the one-dimensional case, we can use a probability density function to define a continuous two-dimensional random variable. If R is a region in the xy-plane, then

$$P((x, y) \text{ is in } R) = \iint_R p(x, y)\, dx\, dy,$$

where the integral is taken over the region R. This integral represents a volume in xyp-space.

Example 2.10 *Computing the probability of a joint event using a two-dimensional probability density function.*

Suppose that (x, y) has the probability density function

$$p(x, y) = \begin{cases} 6(1 - x - y) & \text{if } 0 \le x,\ 0 \le y, \text{ and } x + y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

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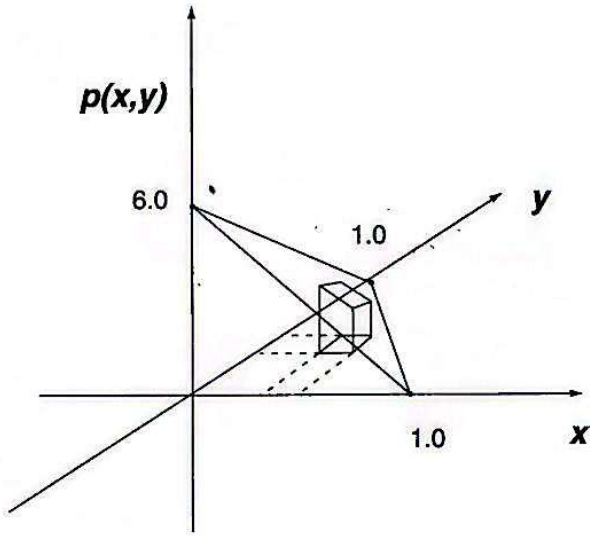


Figure 2.16: The two-dimensional density from Example 2.10.

as shown in Figure 2.16. This qualifies as a density function since it is a pyramid with base area $A = 1/2$, height $h = 6$, and volume $V = (1/3)Ah = 1$. To find the probability of the joint event J, $0.2 \leq x \leq 0.3$ and $0.4 \leq y \leq 0.5$, the density must be integrated over this region:

$$\begin{aligned} P(J) &= \int_{0.2}^{0.3} \left[\int_{0.4}^{0.5} 6(1-x-y)\,dy \right] dx \\ &= \int_{0.2}^{0.3} 6[(0.5-0.4)-(0.5x-0.4x)-(0.5^2-0.4^2)/2]\,dx = 0.018. \end{aligned}$$

The Bivariate Normal Density

Probably the most used joint two-dimensional density is the **bivariate normal density** given by

$$p(x,y) = \frac{e^{-\frac{1}{2(1-\rho_{xy}^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - \frac{2\rho_{xy}(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]}}{2\pi\sigma_x\sigma_y\sqrt{(1-\rho_{xy}^2)}}. \tag{2.21}$$

The three-dimensional graph for a particular example of this density is shown in Figure 2.17a. The contours of constant probability density $p(x,y) = c$ always form ellipses in the xy-plane as shown in Figure 2.17b.

The parameters μ_x and σ_x are called the **mean** and **standard deviation** of the random variable x, and μ_y and σ_y are called the **mean** and **standard deviation** of

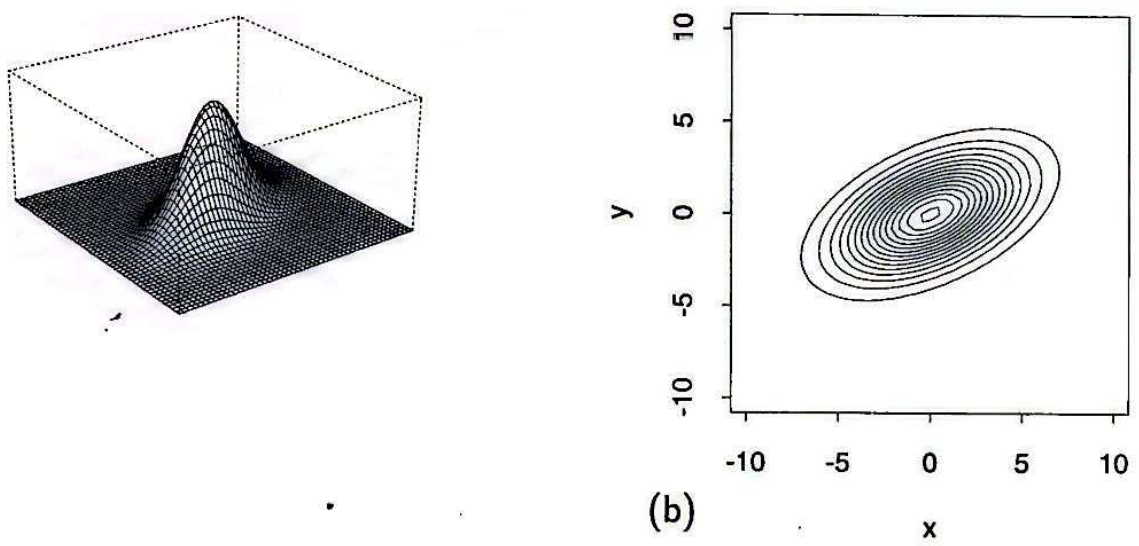


Figure 2.17: A bivariate normal density where $\mu_x = 0$, $\mu_y = 0$, $\sigma_x = 3$, $\sigma_y = 2$, and $\rho_{xy} = 1/2$. (a) A three-dimensional plot. (b) Some contours of constant probability density.

y. The parameters μ_x and μ_y denote the center of the density in the xy-plane, and σ_x and σ_y determine the spread in each direction. The fifth parameter ρ_{xy} is called the **coefficient of linear correlation** or simply the **correlation** between x and y. It influences the orientation of the density with respect to the coordinate axes. These parameters are discussed further in the next section, and classification procedures that use the bivariate normal density are discussed in Chapter 3.

2.5 Moments of Random Variables

If x is a discrete random variable, the **expected value** of x, denoted by $\mathrm{E}(x)$, is defined by

$$\mathrm{E}(x) = \sum_{i=1}^{n} x_i P(x_i) = \mu_x,$$

where $x_1, \ldots, x_n$ are all the possible outcomes of x and $P(x_i)$ is the probability that $x = x_i$. The expected value of x is thus a weighted average of its possible values, each value weighted by the probability of its occurrence. If x has a countably infinite range, then the summation is from $i = 1$ to ∞. The expected value $\mathrm{E}(x)$ is also called the **mean** of x and is also denoted by μ or μ_x.

$\mathrm{E}(x)$ is also called the **first moment** of the distribution. The k**th moment** is defined as

$$\mathrm{E}(x^k) = \sum_{i=1}^{n} x_i^k P(x_i).$$

Thus the second and third moments of a distribution $P(x)$ are the expected values of x^2 and x^3 for that distribution. The **kth central moment**, also called the kth moment about the mean, is defined as

$$\mathrm{E}[(x-\mu_x)^k] = \sum_{i=1}^{n}(x_i-\mu_x)^k P(x_i).$$

The **average value** of x in a randomly sampled data set $x_1, \ldots, x_n$ is defined as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

which is discussed more fully in Section 2.6. This could also be called the mean of the data set. The average value $\bar{x}$ of the data would precisely equal the mean μ of the population from which the data were sampled only if the data set were exactly representative of the probability distribution of the population. In general, the expected difference between $\bar{x}$ and μ approaches 0 as the sample size increases.

If x is a continuous random variable, the **expected value** of x is

$$\mathrm{E}(x) = \int_{-\infty}^{\infty} xp(x)\,dx = \mu_x$$

where $p(x)$ is the density of x. The expected value $\mathrm{E}(x)$ is also denoted by μ_x.

If f is a function of x, then the expected value of $f(x)$ is defined as

$$\mathrm{E}(f(x)) = \sum_{k=1}^{n} f(x_i)P(x_i)$$

when x is discrete. This is the sum of all the possible values of $f(x_i)$, each weighted by its probability of occurrence. When x is continuous, each value of $f(x)$ is weighted by the density $p(x)$ and the product is integrated

$$\mathrm{E}(f(x)) = \int_{-\infty}^{\infty} f(x)p(x)\,dx.$$

If the distribution function or density of a random variable is symmetric about some point c, then that point is the mean of the random variable (Problem 2.51).

Example 2.11 *The expected value for the number of heads when tossing a coin twice.*

When a fair coin is tossed two times, the number of heads obtained x has the probability distribution

x	$P(x)$
0	1/4
1	1/2
2	1/4

The expected value of x is then

$$\mathrm{E}(x) = \sum_{x=0}^{2} xP(x) = 0(1/4) + 1(1/2) + 2(1/4) = 1.$$

On the average, one head will be obtained in two tosses.

Example 2.12 *The expected value of a uniform random variable.*

If x has a uniform density on the interval $[a, b]$, its density function is $p(x) = 1/(b-a)$ on the interval $[a, b]$, and 0 outside of that interval. Thus its expected value is

$$\mathrm{E}(x) = \int_{-\infty}^{\infty} xp(x)\,dx = \int_a^b x\frac{1}{b-a}\,dx = \frac{1}{b-a}\,\frac{x^2}{2}\bigg|_a^b = \frac{a+b}{2},$$

which is, as expected, at the center of its range.

Example 2.13 *The expected value of an exponential random variable.*

The expected value of an exponential random variable t is expressed by

$$\mathrm{E}(t) = \int_0^{\infty} t\beta e^{-\beta t}\,dt.$$

Using integration by parts with $u = t$ and $dv = \beta e^{-\beta t}dt$, we find that

$$\mathrm{E}(t) = \frac{t\beta e^{-\beta t}}{-\beta}\bigg|_0^{\infty} - \int_0^{\infty} \frac{\beta e^{-\beta t}}{-\beta}\,dt = 0 + \int_0^{\infty} e^{-\beta t}\,dt = \frac{e^{-\beta t}}{-\beta}\bigg|_0^{\infty} = \frac{1}{\beta}.$$

Thus if $\beta = 0.01$, the expected or average lifetime is 100. However, the half-life is 69.31 (see Example 2.7).

The second moment of a random variable x is defined to be $\mathrm{E}(x^2)$, but the **second central moment** or **variance** of a random variable x is defined by

$$\sigma_x^2 = \mathrm{E}[(x - \mathrm{E}(x))^2] = \mathrm{E}[(x - \mu_x)^2]. \quad = \sum_{i}^{n} (x_i - \mu_x)^2 P(x_i) \quad (2.22)$$

or

$$= \int (x - \mu_x)^2 p(x)\,dx$$

This is the expected squared deviation of x from its mean and describes the spread of the values of the random variable x about its mean. The variance is a convenient measure of the spread of a distribution, but it has the disadvantage that it is measured in squared units. In order to have a measure of the spread that is measured in the units of the original random variable, one must use the **standard deviation** σ_x which is defined as the square root of the variance σ_x^2.

An important property of the expectation operator $\mathrm{E}(\cdot)$ is that it is a **linear operator** which means that if x and y are random variables, and z is a new random variable which is the weighted sum $ax + by$, then

$$\mathrm{E}(z) = \mathrm{E}(ax + by) = a\mathrm{E}(x) + b\mathrm{E}(y). \tag{2.23}$$

To see this in the continuous case, let $p(x, y)$ be the joint probability distribution of (x, y). Also let $p_1(x) = \int_{-\infty}^{\infty} p(x, y)\, dy$ and $p_2(y) = \int_{-\infty}^{\infty} p(x, y)\, dx$ be the **marginal densities** of x and y respectively. (The **marginal density** of x is the density of the component x in (x, y) regardless of the outcome of y.) Then

$$\begin{aligned}
\mathrm{E}(z) &= \mathrm{E}(ax + by) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (ax + by)p(x, y)\, dx\, dy \\
&= a\int_{-\infty}^{\infty} x\left[\int_{-\infty}^{\infty} p(x, y)\, dy\right] dx + b\int_{-\infty}^{\infty} y\left[\int_{-\infty}^{\infty} p(x, y)\, dx\right] dy
\end{aligned}$$

Now since

$$\int_{-\infty}^{\infty} p(x, y)\, dx = p_2(y) \quad \text{and} \quad \int_{-\infty}^{\infty} p(x, y)\, dy = p_1(x),$$

$$\mathrm{E}(ax + by) = a\int_{-\infty}^{\infty} xp_1(x)\, dx + b\int_{-\infty}^{\infty} yp_2(y)\, dy = a\mathrm{E}(x) + b\mathrm{E}(y).$$

The proof in the discrete case is done by replacing integrals and densities by summations and distribution functions:

$$\begin{aligned}
\mathrm{E}(z) &= \mathrm{E}(ax + by) = \sum_x \sum_y (ax + by)P(x, y) \\
&= a\sum_x \sum_y xP(x, y) + b\sum_x \sum_y yP(x, y) \\
&= a\sum_x x\sum_y P(x, y) + b\sum_y y\sum_x P(x, y) \\
&= a\sum_x xP_1(x) + b\sum_y yP_2(y) \\
&= a\mathrm{E}(x) + b\mathrm{E}(y).
\end{aligned}$$

Because $\mathrm{E}(\cdot)$ is a linear operator, we can obtain an alternative expression of the variance (compare with (2.22)):

$$\sigma_x^2 = \mathrm{E}[(x - \mathrm{E}(x))^2] = \mathrm{E}[x^2 - 2x\mathrm{E}(x) + \mathrm{E}(x)^2]$$

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$$= \mathrm{E}(x^2) - 2\mathrm{E}[x\mathrm{E}(x)] + \mathrm{E}[\mathrm{E}(x)^2].$$

But $\mathrm{E}(x) = \mu_x$ is a constant, so

$$\mathrm{E}[x\mathrm{E}(x)] = \mathrm{E}(x\mu_x) = \mu_x\mathrm{E}(x) = \mu_x^2$$

and

$$\sigma_x^2 = \mathrm{E}(x^2) - 2\mu_x^2 + \mu_x^2 = \mathrm{E}(x^2) - \mu_x^2. \tag{2.24}$$

This form is more convenient than (2.22) for computing the variance of a data set.

Example 2.14 *The variance of a uniform random variable with range a to b.*

Between a and b the density is $1/(b-a)$ so the second moment is

$$\mathrm{E}(x^2) = \int_a^b x^2(1/(b-a))\,dx = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}.$$

Using (2.24),

$$\sigma_x^2 = \mathrm{E}(x^2) - \mu_x^2 = \frac{b^2 + ab + a^2}{3} - \left(\frac{b+a}{2}\right)^2 = \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}.$$

Example 2.15 *Find the variance of the number of heads obtained from two coin flips.*

By using the distribution function from Example 2.11, and the value $\mu_x = 1$, we obtain

$$\begin{aligned} \sigma_x^2 &= E(x-\mu_x)^2 = \sum_{x=0}^{2}(x-1)^2 P(x) \\ &= (-1)^2(1/4) + 0^2(1/2) + 1^2(1/4) = 1/2. \end{aligned}$$

This result could also be obtained using (2.24).

Example 2.16 *The mean of a Poisson random variable.*

The mean of a Poisson random variable is

$$\mu = \mathrm{E}(x) = \sum_{x=0}^{\infty} xP(x) = \sum_{x=1}^{\infty} xP(x)$$

$$= \sum_{x=1}^{\infty} x\frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\sum_{x=1}^{\infty}\frac{x\lambda^x}{x!} = \lambda e^{-\lambda}\sum_{x=1}^{\infty}\frac{\lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^x}{x!}$$

Since the Maclaurin series expansion of e^λ is

$$e^\lambda = \sum_{x=0}^{\infty}\frac{\lambda^x}{x!},$$

$$\mu = \lambda e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^x}{x!} = \lambda e^{-\lambda}e^\lambda = \lambda.$$

Example 2.17 *The mean and variance of the standard normal distribution.*

In this example, we show that, for a standard normal random variable x with density

$$p(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2},$$

$\mathrm{E}(x) = 0$ and $\mathrm{E}(x^2) = 1$. We calculate

$$\begin{aligned}\mathrm{E}(x) &= \int_{-\infty}^{\infty} xp(x)\,dx \\ &= \int_{-\infty}^{\infty} x\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\,dx = 0.\end{aligned} \tag{2.25}$$

Since the integrand $f(x) = xe^{-x^2/2}/\sqrt{2\pi}$ satisfies $f(-x) = -f(x)$, its integral over the entire range is zero.

To show that $\mathrm{E}(x^2) = 1$, we use integration by parts:

$$\int u\,dv = uv - \int v\,du,$$

where $u = x$, $dv = x(1/\sqrt{2\pi})e^{-x^2/2}\,dx$, $du = dx$, $v = -(1/\sqrt{2\pi})e^{-x^2/2}$, and

$$\begin{aligned}\mathrm{E}(x^2) &= \int_{-\infty}^{\infty} x^2\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\,dx \\ &= -x\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\Big|_{-\infty}^{\infty} - \left(-\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\,dx\right) \\ &= 0 - (-1) = 1.\end{aligned}$$

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L'Hôpital's rule shows that the first term in the expression for $\mathrm{E}(x^2)$ is zero at each limit. The second term is the integral of the normal density and is equal to 1 since the area under a probability density function must be 1 (see Problem 2.29).

Example 2.18 *The mean and variance of a normal distribution.*

To justify the definitions of the terms mean and variance for the parameters μ and σ^2 for the normal distribution in Section 2.3, we must verify that the expected value for a normal distribution is actually the parameter μ and that the variance is the parameter σ^2. By making the substitution $z = (x-\mu)/\sigma$ or $x = \sigma z + \mu$, and $dz = dx/\sigma$, we can use the results from Example 2.17 to obtain

$$\begin{aligned}
\mathrm{E}(x) &= \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx \\
&= \int_{-\infty}^{\infty} (\sigma z + \mu) \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz \\
&= \sigma \mathrm{E}(z) + \mu \mathrm{E}(1) = \sigma \cdot 0 + \mu \cdot 1 = \mu.
\end{aligned}$$

Also,

$$\begin{aligned}
\mathrm{E}(x^2) &= \int_{-\infty}^{\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx \\
&= \int_{-\infty}^{\infty} (\sigma z + \mu)^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz \\
&= \sigma^2 \mathrm{E}(z^2) + 2\sigma\mu \mathrm{E}(z) + \mu^2 \mathrm{E}(1) \\
&= \sigma^2 \cdot 1 + 2\sigma\mu \cdot 0 + \mu^2 \cdot 1 = \sigma^2 + \mu^2.
\end{aligned}$$

The variance of x is $\mathrm{E}(x^2) - \mathrm{E}(x)^2 = (\sigma^2 + \mu^2) - \mu^2 = \sigma^2$, so we were justified in calling μ and σ^2 the mean and variance of the normal density.

Moments of Bivariate Distributions

Moments can also be computed for bivariate distributions. If (x, y) is a joint random variable, then μ_x and σ_x are the mean and standard deviation of x, and μ_y and σ_y are the mean and standard deviation of y. The **covariance** of x and y is defined as

$$\sigma_{xy} = \mathrm{E}[(x - \mu_x)(y - \mu_y)].$$

If x and y for a sample are both greater than or both less than their means, the product $(x-\mu_x)(y-\mu_y)$ will be positive. Thus the covariance indicates how much x and y tend to vary together. The value of the covariance depends on how much each variable tends to deviate from its mean, as represented by the standard deviations of x and y, and also depends on the degree of association between x and y. To eliminate the scaling effect of σ_x and σ_y, we can divide the covariance by the product of the two standard deviations to yield the **correlation** between x and y, which is usually denoted by the Greek letter ρ (rho):

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \mathrm{E}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)\right].$$

It is also called the coefficient of linear correlation, or **correlation coefficient**. While the covariance of x and y, σ_{xy}, has units equal to the product of the units of x and of y, the correlation is a dimensionless quantity. The correlation has the convenient property that for any random variables x and y

$$-1 \leq \rho_{xy} \leq 1.$$

If ρ_{xy} is positive, x and y are said to be positively correlated. If $\rho = 1$, they are perfectly correlated, and y is a linear function of x. If ρ_{xy} is negative, x and y are negatively correlated or "anticorrelated." If the correlation is 0, x and y are said to be uncorrelated. The correlation is symmetric in the sense that $\rho_{xy} = \rho_{yx}$.

If a random variable z is a weighted sum of the random variables x and y, the variance of z depends not only on the variances of x and y but also on the covariance of x and y. Let $z = ax + by$. The variance σ_z^2 of z is

$$\begin{aligned}
\mathrm{E}[(z-\mu_z)^2] &= \mathrm{E}[((ax+by-(a\mu_x+b\mu_y))^2] \\
&= a^2\mathrm{E}(x^2 - 2x\mu_x + \mu_x^2) + 2ab\mathrm{E}(xy - x\mu_y - y\mu_x + \mu_x\mu_y) \\
&\quad + b^2\mathrm{E}(y^2 - 2y\mu_y + \mu_y^2) \\
&= a^2\sigma_x^2 + 2ab\sigma_{xy} + b^2\sigma_y^2,
\end{aligned}$$

using (2.23). As a special case, when $\sigma_{xy} = 0$,

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2. \tag{2.26}$$

The Multivariate Normal Density

The normal density can be generalized to any number of dimensions. When there are more than two dimensions, using matrix notation to simplify the expressions is helpful. The joint random variable $(x_1, \ldots, x_d)$ is called a d-dimensional **random vector** and will be denoted by $\mathbf{x}$. Let $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)$ be the vector of means of $\mathbf{x}$, and let σ_{ij} be

the covariance of x_i and x_j. (Note that the covariance σ_{ii} of a random variable with itself is actually the variance σ_i^2.)

The components of a random vector are often written in the column vector form:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix}.$$

We define the **mean vector** as

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{x}) = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_d \end{pmatrix}. \tag{2.27}$$

The **covariance matrix** is symmetric ($\sigma_{ij} = \sigma_{ji}$) and is usually denoted by a capital sigma:

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{21} & \cdots & \sigma_{d1} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{d2} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1d} & \sigma_{2d} & \cdots & \sigma_{dd} \end{pmatrix} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1d} & \sigma_{2d} & \cdots & \sigma_d^2 \end{pmatrix}. \tag{2.28}$$

The d-dimensional normal density can be written using matrix notation:

$$p(\mathbf{x}) = \frac{1}{\sqrt{\det \boldsymbol{\Sigma} (2\pi)^d}} e^{-\frac{(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}{2}} \tag{2.29}$$

where det denotes the determinant. The exponent in (2.29) can be expressed algebraically as

$$\frac{1}{2} \sum_{ij} (x_i - \mu_i) s_{ij} (x_j - \mu_j)$$

where s_{ij} is the ijth component of $\boldsymbol{\Sigma}^{-1}$, the inverse of the covariance matrix Σ.

When $d = 2$,

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix},$$

and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix} = \begin{pmatrix} \sigma_x^2 & \rho_{xy}\sigma_x\sigma_y \\ \rho_{xy}\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix},$$

so we obtain (2.21). Examples of classification procedures which use the multivariate normal density will be discussed in Chapter 3.

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2.6 Estimation of Parameters from Samples

To use the densities discussed in this chapter for the parametric classification procedures described in Chapter 3, some method must be available for estimating the parameters of the density from the data available in the training sample. We discuss three kinds of estimates for these parameters: method of moments estimates, maximum likelihood estimates, and unbiased estimates. Other desirable properties of estimates are discussed in many statistics texts.

The Method of Moments

If some particular family of densities (or probability distributions) is assumed to describe a data set, one way to choose the r adjustable parameters in the density is such that the first r central moments of the density will equal the first r central moments of the data. To estimate parameters by the **method of moments**, n independent samples or patterns $x_1, x_2, \ldots, x_n$ are collected from the random variable x, which may be continuous or discrete. If we randomly choose one of these samples in the data set, we can consider its value to be a new discrete random variable x' called an **empirical random variable**, which takes on one of the values $x_1, x_2, \ldots, x_n$, each with probability $1/n$. (These values need not necessarily be distinct.) The **sample mean** is defined by

$$\bar{x} = \mathrm{E}(x') = \sum_{i=1}^{n} x_i P(x_i) = \frac{1}{n} \sum_{i=1}^{n} x_i,$$

which is the average of all the samples. The **sample variance** is the variance of x' and is denoted in this section by $\sigma_{x'}^2$. It is given by

$$\sigma_{x'}^2 = \sum_{i=1}^{n} (x_i - \mathrm{E}(x'))^2 P(x_i) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

If the mean and variance are the density function parameters to be estimated, the sample mean and the sample variance are simply used as the estimate for the population. If other parameters are required, they are found from the moments as in Example 2.19. If the distribution contains more than two parameters to be estimated, higher moments such as $E(x' - \bar{x})^3$ and $E(x' - \bar{x})^4$ are also used.

The method of moments can also be used to estimate the covariance σ_{xy} of a bivariate distribution. We compute the covariance of the sample

$$\sigma_{x'y'} = \mathrm{E}[(x' - \mathrm{E}(x'))(y' - \mathrm{E}(y'))] = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}),$$

which is then used as an estimate of the covariance of the population from which the sample was chosen.

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Example 2.19 *The method of moments estimate of the range of a uniform density.*

Suppose we know that the sample values $x = 2, 3, 5, 6, 8, 9, 11$, and 12 come from a uniform distribution, but we do not know the values of the parameters a and b that determine the range. To estimate these parameters using the method of moments, we compute

$$\bar{x} = 7 \quad \text{and} \quad \sigma_{x'}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 11.5.$$

A uniform density on the interval $[a, b]$ has mean $\mu = (a + b)/2$ and variance $\sigma^2 = (b - a)^2/12$ (see Examples 2.12 and 2.14). Thus we estimate a and b by $\hat{a}$ and $\hat{b}$. (In general, a "hat" over a parameter denotes an estimate of that parameter.)

$$\frac{\hat{a} + \hat{b}}{2} = 7 \quad \text{and} \quad \frac{(\hat{b} - \hat{a})^2}{12} = 11.5.$$

Solving for $\hat{a}$ and $\hat{b}$, we obtain

$$\hat{a} = 1.13 \quad \text{and} \quad \hat{b} = 12.87.$$

Thus a uniform distribution with a range of 1.13 to 12.87 has the same mean and variance as our data, so the method of moments estimate of the range of the distribution is $(1.13, 12.87)$.

A disadvantage of using the method of moments to estimate the range of a uniform distribution is that some samples may lie outside the estimated range $[\hat{a}, \hat{b}]$ (see Problem 2.56), in which case the estimate is logically inconsistent with the data.

Maximum Likelihood Estimates

Although the method of moments is easy to understand and works well in many cases, there are other methods for choosing parameters that are also appealing. The **maximum likelihood estimate** of a parameter is that value which, when substituted into the probability distribution (or density if the random variable is continuous), produces that distribution for which the probability (or joint density) of obtaining the entire observed set of samples is maximized. To compute the maximum likelihood estimate, we choose the parameter (or set of parameters) that maximizes the value of the joint distribution function or multivariate density function for the entire data set when it is evaluated at the set of sample points $x_1, \ldots, x_n$.

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MLE: 1. Take ln of function
2. Take derivative of ① wrt variable
3. Set equal to zero
4. Solve for variable

Example 2.20 *The maximum likelihood estimate of the probability of getting a head when flipping a biased coin.*

Consider the discrete random variable x for which $P(H) = \theta$, and $P(T) = 1 - \theta$. To estimate the value of θ, obtain n independent outcomes $x_1, x_2, \ldots, x_n$ from x. Because the outcomes are independent, the probability of this set of data is

$$P(x_1, \ldots, x_n) = P(x_1)P(x_2)\ldots P(x_n) = \theta^k(1-\theta)^{n-k} \tag{2.30}$$

where θ is the probability of obtaining a head, k is the number of heads obtained, and n is the number of samples collected. The maximum likelihood estimate for θ is the value that maximizes (2.30).

To simplify the calculations, we note that the value of a parameter that maximizes an expression is the same as the value that maximizes the logarithm of the expression. The logarithm of (2.30) is

$$\ln P(x_1, \ldots, x_n) = k \ln \theta + (n-k)\ln(1-\theta).$$

Taking the derivative with respect to θ and setting the result to zero gives

$$\frac{k}{\theta} - \frac{n-k}{1-\theta} = 0$$

$k - k\theta - [n\theta - k\theta] = 0$
$k - k\theta - n\theta + k\theta = 0$
$k - n\theta = 0 \quad \theta = k/n$

or $\theta = k/n$. We note that this estimate is the same as the sample mean which is also the method of moments estimate.

Example 2.21 *The maximum likelihood estimate of the range of a uniform distribution.*

The maximum likelihood estimate $\hat{a}$ for parameter a of a uniform distribution with range $[a, b]$ is the smallest sample value found among the independent samples x_1, x_2, $\ldots$, x_n. The maximum likelihood estimate $\hat{b}$ for b is the largest sample value. Any narrower range for $[\hat{a}, \hat{b}]$ would be impossible, given the data, and any larger range would reduce the joint density function $p(x_1, \ldots, x_n) = [1/(b-a)]^n$ that produced the samples. For example, if the samples are $x_1 = 9.2$, $x_2 = 2.1$, $x_3 = 3.7$, $x_4 = 11.1$, and $x_5 = 9.2$, the maximum likelihood estimates are $\hat{a} = 2.1$ and $\hat{b} = 11.1$.

Although this estimate is consistent with the data, the true range $[a, b]$ from which the data came is probably larger than the range $[\hat{a}, \hat{b}]$ of the actual data obtained, since it is extremely unlikely that the limiting values a and b actually occurred in the data set. The actual range is probably larger than the estimated range. Because of this, the maximum likelihood estimate is said to be a **biased** estimate of the range of a uniform density.

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Example 2.22 *Finding the maximum likelihood estimate for μ in a normal distribution.*

The density of a normal random variable with mean μ and variance σ^2 is

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

If all the random samples $x_1, x_2, \ldots, x_n$ are independent,

$$p(x_1, \ldots, x_n) = p(x_1) \ldots p(x_n) = \prod_{i=1}^{n} p(x_i),$$

where $\prod$ represents taking the product of the terms that follow it. Therefore

$$p(x_1, \ldots, x_n) = p(x_1) \ldots p(x_n) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}. \tag{2.31}$$

The derivative of the logarithm of (2.31) with respect to μ is

$$\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)\cdot 2 = \frac{1}{\sigma^2}\left[\sum_{i=1}^{n} x_i - n\mu\right].$$

Setting this derivative equal to 0 and solving for μ results in the maximum likelihood estimate for μ:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}.$$

This estimate for μ is the same as the method of moments estimate.

Example 2.23 *The maximum likelihood estimate for σ^2 in a normal distribution.*

To simplify the notation, we replace σ^2 by v in (2.31) to obtain

$$p(x_1, \ldots, x_n) = \frac{1}{v^{n/2}(2\pi)^{n/2}} e^{-\frac{1}{2v}\sum_{i=1}^{n}(x_i-\mu)^2}. \tag{2.32}$$

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The maximum likelihood estimate of v is the value that maximizes (2.32), so we differentiate its logarithm with respect to v to obtain

$$\frac{d}{dv}\ln p(x_1,\ldots,x_n) = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

Setting this to zero and solving for v produces

$$\hat{\sigma}^2 = \hat{v} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2,$$

so the maximum likelihood estimate of the variance equals the variance of the sample if μ is known. If μ and σ^2 are both unknown, setting the partial derivatives of (2.31) with respect to μ and σ^2 to 0 simultaneously yields the same solutions. This is the same estimate for σ^2 as was obtained using the method of moments.

Although it will not be proven here, it turns out that when d features are multivariate normally distributed, the means, variances and covariances of the sampled data are maximum likelihood estimates of the means, variances, and covariances of the population.

The sample mean $\bar{x}$ is the maximum likelihood estimate for the mean μ of a normal distribution, as shown in Example 2.22, but this is not the case for all distributions. The sample median, not the sample mean, is the maximum likelihood estimate of the center of the double exponential distribution. In the case of the Cauchy distribution, the maximum likelihood estimate for a in (2.20) is neither the sample mean nor the sample median but must be obtained numerically (see Problem 2.46). Since the Cauchy density has an infinite variance, the method of moments cannot be used to estimate its spread parameter b. The following example shows that a maximum likelihood estimate can be quite counterintuitive. Compare this example with the unbiased estimate described in Example 2.25.

Example 2.24 *The maximum likelihood estimate for estimating the number of cards in a numbered deck, given only one draw.*

Consider a well-shuffled deck of N cards which are labeled $1, 2, \ldots, N$. The label of a single card drawn from this deck is a discrete random variable x, where the probability of any particular value is $1/N$; that is, $P(1) = 1/N$, $P(2) = 1/N$, ..., $P(N) = 1/N$. If we draw one card, we know that N is at least as large as the value x we drew. If $N = x$, the probability of drawing x would be $1/x$. If N is larger than x, say $N = x+1$, then

the probability of obtaining x would be less, say $P(x) = 1/(x+1)$, which is smaller than $1/x$. Thus the distribution that has the highest probability of producing x is the distribution for which $N = x$. The following table shows the probability of drawing a card x for various assumed values of N.

N	$P(x)$
1	0
2	0
⋮	⋮
$x-1$	0
x	$1/N$
$x+1$	$1/(N+1)$
$x+2$	$1/(N+2)$
⋮	⋮

The probability $P(x)$ is maximized when $x = N$, so the maximum likelihood estimate of N is $\hat{N} = x$. (If several cards had been drawn, their largest value would be the maximum likelihood estimate of N.)

The maximum likelihood estimate x may not seem to be a good guess for N, because N can never be less than x and is probably higher because it seems unlikely that we drew the highest card in the deck. This means that this maximum likelihood estimate is biased in the direction of choosing a value for $\hat{N}$ that is less than the true value. It is plausible that there are as many cards higher than the card x we drew as there are cards that are lower than x. This reasoning leads to the concept of an unbiased estimate, is discussed in the next section.

Unbiased Estimators

Another desirable property for an estimate $\hat{\theta}$ of θ is that it be **unbiased**. An unbiased estimate $\hat{\theta}$ is an estimate for θ that satisfies

$$\mathrm{E}(\hat{\theta}) = \theta.$$

An unbiased estimate is neither too low or too high on the average, because the expected value of the estimate equals the parameter being estimated.

Notice that since μ_x is defined as $\mathrm{E}(x)$,

$$\mathrm{E}(\bar{x}) = \mathrm{E}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}(x_i) = \frac{1}{n}(n\mu_x) = \mu_x.$$

Thus $\bar{x}$ is always an unbiased estimate of μ_x, regardless of the type of distribution or density.

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Example 2.25 *Obtaining an unbiased estimate of the number of cards in a numbered deck from only one sample.*

As in Example 2.24, the probability of choosing any particular card x from a deck containing N cards labeled $1, \ldots, N$ is $1/N$. If the drawn card x was the average card in the deck, there are $x-1$ cards with smaller numbers and $x-1$ cards with larger numbers, so the total number of cards in the deck is $2(x-1)+1 = 2x-1$. We will show that $\hat{N} = 2x-1$ is an unbiased estimate for N. To be unbiased, the expected value of an estimate must equal the parameter value being estimated, which in this case means $\mathrm{E}(\hat{N}) = N$. We have

$$\mathrm{E}(\hat{N}) = \mathrm{E}(2x-1) = 2\mathrm{E}(x) - 1 =$$

$$2\sum_{x=1}^{N} x \cdot \frac{1}{N} - 1 = \frac{2}{N} \cdot \frac{N(N+1)}{2} - 1 = N$$

which verifies that $\hat{N} = 2x-1$ is an unbiased estimate for N. If $N = 2x-1$, then there are as many cards higher than x as there are lower than x. If several cards were drawn, $\hat{N} = 2\bar{x}-1$ would be an unbiased estimate of N, as can be seen by substituting $\bar{x}$ for x in the preceding equations. As the number of cards drawn becomes large, the difference between the maximum likelihood estimate and the unbiased estimate becomes small. If *all* of the cards were drawn, these estimates would be identical, because the average of all the card numbers from 1 to N would be $(N+1)/2$, so $N = 2\bar{x} - 1$, and the largest number drawn would also equal N.

The maximum likelihood estimate $\hat{N} = x$ obtained for one draw in Example 2.24 is very different from the unbiased estimate $\hat{N} = 2x-1$. The average value of a large number of independent unbiased estimates of a parameter is expected to approach the true value of the parameter, while the average of a large number of maximum likelihood estimates of a parameter may be extremely biased. Most people prefer the unbiased estimate because it does not consistently overestimate or underestimate the value of a parameter.

Unlike method of moments and maximum likelihood estimators, there is no single systematic method for obtaining an unbiased estimate for an arbitrary parameter. Unbiased estimates are usually created by correcting for the bias in another estimate, such as a method of moments or maximum likelihood estimator. The bias is found by computing the expected value of the estimator and comparing it to the true value of the parameter.

For example, if one is given a single sample x from a uniform density with range $[0, b]$, what is the maximum likelihood estimator of b, based on x? Using reasoning

similar to Example 2.24, the maximum likelihood estimator for b is x. This estimate is clearly biased. To determine the bias, we calculate

$$\mathrm{E}(x) = \int_0^b xp(x)\,dx = \int_0^b x\,\frac{dx}{b} = \frac{b}{2}.$$

Thus, on the average, the maximum likelihood estimate x would only be half of the true parameter value b. To remove the bias, we double the maximum likelihood estimator: $\hat{b} = 2x$ is an unbiased estimate of b.

Now consider two samples x_1 and x_2 drawn from a uniform distribution with unknown range $[a, b]$. The maximum likelihood estimator of $b-a$ is $|x_1 - x_2|$ (see Problem 2.48). From Problem 2.49, an unbiased estimate of $b - a$ is three times the maximum likelihood estimate. On the average, the two samples divide the density into three equal areas. In general, n samples would divide the density into $n + 1$ equal areas on the average, which is the reason for choosing the expected z-values as they were in the normal plots in Section 2.6.

In general there are many possible unbiased estimates for a parameter. For example, any weighted average of the original samples

$$\hat{\mu} = \sum_{i=1}^{n} w_i x_i \quad \text{where} \quad \sum_{i=1}^{n} w_i = 1 \tag{2.33}$$

is an unbiased estimate for μ because

$$\mathrm{E}(\hat{\mu}) = \mathrm{E}\left(\sum_{i=1}^{n} w_i x_i\right) = \sum_{i=1}^{n} w_i \mathrm{E}(x_i) = \sum_{i=1}^{n} w_i \mu = \mu.$$

The average of the first and last samples drawn, $(x_1 + x_n)/2$, and even the first sample value x_1, are unbiased estimates for μ; but these estimates would have a much greater expected error than would the estimate $\bar{x}$. In fact, $\bar{x}$ is the best unbiased linear estimate for μ in the sense that $\bar{x}$ has the smallest variance of all unbiased estimates of μ which are weighted averages of the sample values of the form (2.33).

We verify this assertion for the simple case of two samples x_1 and x_2 by showing that the variance of $ax_1 + (1 - a)x_2$ is minimized when $a = 0.5$. Since x_1 and x_2 are independent, (2.26) indicates that the variance of $ax_1 + (1 - a)x_2$ is $[a^2 + (1 - a)^2]\sigma^2$ where σ^2 is the variance of x_1 or x_2. Differentiating with respect to a and setting the result to zero produces $[2a - 2(1 - a)]\sigma^2 = 0$, or $a = 0.5$.

An Unbiased Estimator for the Variance

Although $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$ is an unbiased estimate for μ, it can be shown that the variance of the sample $\sigma_{x_i}^2 = (1/n)\sum_{i=1}^{n}(x_i - \bar{x})^2$ is not an unbiased estimator for the

variance of the population σ^2. In fact, no matter from which distribution the samples are drawn,

$$\mathrm{E}(\sigma_{x'}^2) = \frac{n-1}{n}\sigma^2. \tag{2.34}$$

Thus, the variance of the data about its mean is expected to be less than the variance of the population about its mean. Bias is introduced because the second moment of the data is taken about the mean of the data, which is known to be smaller than the second moment about any other point. In particular, it is expected to be smaller than the second moment about the true mean of the population. (If the true mean of the population were known, the second moment of the data about the population mean would be an unbiased estimate of the variance of the population.) A proof of (2.34) is outlined in Problem 2.50.

Multiplying (2.34) by $n/(n-1)$ removes the bias:

$$\begin{aligned}
\frac{n}{n-1}\mathrm{E}\left(\sigma_{x'}^2\right) &= \mathrm{E}\left(\frac{n}{n-1}\sigma_{x'}^2\right) \\
&= \mathrm{E}\left[\frac{n}{n-1}\cdot\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right] \\
&= \mathrm{E}\left[\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2\right] \\
&= \frac{n}{n-1}\cdot\frac{n-1}{n}\sigma^2 = \sigma^2.
\end{aligned}$$

Thus, dividing by $n-1$ instead of n produces an unbiased estimate s^2 of the variance of the population:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2. \tag{2.35}$$

In practice, the unbiased estimate s^2 is usually used instead of the maximum likelihood estimator $\sigma_{x'}^2$. The standard statistical tables such as the chi-squared, t, and F tables also assume that the unbiased estimate of σ^2 is used.

In the multivariate case,

$$\bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_d \end{pmatrix}$$

is an unbiased estimate for

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_d \end{pmatrix}$$

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and

$$\mathbf{S} = \begin{pmatrix} s_{11} & \cdots & s_{1d} \\ \vdots & \ddots & \vdots \\ s_{1d} & \cdots & s_{dd} \end{pmatrix}$$

is an unbiased estimate for

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \cdots & \sigma_{1d} \\ \vdots & \ddots & \vdots \\ \sigma_{1d} & \cdots & \sigma_{dd} \end{pmatrix},$$

where

$$s_{jk} = \frac{1}{n-1} \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

The estimated correlation ρ_{jk} is the same regardless of which type of estimate of σ_{jk} and σ_j^2 is used because the factors n or $n-1$ cancel.

2.7 Minimum Risk Estimators

Consider the following problem: Predict the next outcome of a random variable x whose probability distribution function $P(x)$ is known. The "best" prediction depends on the penalty or loss if the prediction is wrong. We will see that the best answer can be the mean, median, or mode of the distribution depending on which loss function is used. In each case, the best prediction is the one that minimizes the **risk**, which is defined to be the expected loss.

Initially we will assume that x is a discrete random variable. One common penalty function is the **squared error loss function**, in which the penalty for guessing the wrong value is equal (or proportional) to the square of the error

$$L(\theta, x) = (x - \theta)^2.$$

Here, $L(\theta, x)$ is the loss incurred for predicting θ when in fact x is the next outcome. The risk of predicting θ is

$$R(\theta) = E[L(\theta, x)] = E[(x - \theta)^2].$$

If x is a discrete random variable,

$$R(\theta) = \sum_{i=1}^{n} (x_i - \theta)^2 P(x_i).$$

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To minimize $R(\theta)$, differentiate it with respect to θ, set the result equal to 0, and solve for θ:

$$\begin{aligned}\frac{d}{d\theta}R(\theta) &= \frac{d}{d\theta}\sum_{i=1}^{n}(x_i-\theta)^2P(x_i)\\ &= -2\sum_{i=1}^{n}(x_i-\theta)P(x_i)=0\end{aligned}$$

or

$$\sum_{i=1}^{n}x_iP(x_i)=\theta\sum_{i=1}^{n}P(x_i).$$

Solving for θ and using the facts that $\sum_{i=1}^{n}P(x_i)=1$ and $\sum_{i=1}^{n}x_iP(x_i)=E(x)$, we obtain

$$\theta=E(x)=\mu_x.$$

Thus the squared error loss function leads to the mean μ_x as the minimum risk predictor θ of the next outcome of x.

Often there is no good reason to square the error to establish a penalty, and it is more reasonable to set the penalty to be proportional to the error. If the penalty is equal (or proportional) to the absolute value of the amount of error, rather than its square, we obtain the **absolute error loss function**

$$L(\theta,x)=|x-\theta|.$$

The risk of choosing θ is

$$R(\theta)=E[L(\theta,x)]=\sum_{i=1}^{n}|x_i-\theta|P(x_i). \tag{2.36}$$

The absolute value terms can be eliminated by breaking this sum into three parts: those for which $x_i>\theta$ ($x_i-\theta$ is positive), those for which $x_i<\theta$ ($x_i-\theta$ is negative), and those for which $x_i=\theta$ ($x_i-\theta$ is zero)

$$R(\theta)=\sum_{x_i>\theta}(x_i-\theta)P(x_i)+\sum_{x_i<\theta}(\theta-x_i)P(x_i)+\sum_{x_i=\theta}0\cdot P(x_i). \tag{2.37}$$

To minimize $R(\theta)$, set its derivative to zero:

$$\frac{dR(\theta)}{d\theta}=-\sum_{x_i>\theta}P(x_i)+\sum_{\theta<x_i}P(x_i)=0.$$

Thus θ should be chosen so that the probability that x_i is greater than θ is equal to the probability that x_i is less than θ, in order to minimize the expected absolute error, if possible. Note that this is a sufficient, but not a necessary, condition and may not

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be possible for an asymmetric discrete distribution. (It is possible for all symmetric distributions.) For example, if $P(0) = 1/3$, $P(1) = 1/2$, and $P(2) = 1/6$, the minimum risk guess is $\theta = 1$, even though $P(x_i > 1) \neq P(x_i < 1)$. Increasing θ above 1 would move it away from $1/3 + 1/2 = 5/6$ of the probability, and decreasing it below 1 moves it away from $1/2 + 1/6 = 2/3$ of the probability, both of which would increase the error. Thus $R(\theta)$ is minimized when θ divides the distribution into two parts that are as nearly equal as possible.

Whenever a distribution can be divided into two halves of equal probability by a value θ, θ is called a **median** of the distribution. The median of a ranked list of numbers is the middle number, if there are an odd number of entries in the list. If the list contains an even number of entries, any value of θ between the two middle numbers can serve as a median. By convention, the median is often taken to be the average of the two middle numbers.

The median is often a more useful statistic than the sample mean. For example, median annual income is more commonly reported than mean annual income. Even though a typical citizen may earn \$10,000 to \$50,000, the sample mean will be strongly influenced by those few individuals with huge annual incomes. The median is more representative of the typical citizen; half of the people earn less than this amount and half earn more.

Example 2.26 *The best estimate using the absolute error loss function.*

Consider the discrete distribution defined by

x	$P(x)$
0	1/4
1	1/4
2	1/4
3	1/4

From (2.36) the risk of choosing θ is

$$R(\theta) = (1/4)|\theta - 0| + (1/4)|\theta - 1| + (1/4)|\theta - 2| + (1/4)|\theta - 3|.$$

Figure 2.18 shows a graph of the risk using the absolute error penalty function versus θ. If $\theta < 0$, $R(\theta)$ can be decreased by moving θ to the right (increasing it) because it will be moving closer to all four of the possible values of x, thus reducing the penalty no matter which of the four values of x occurs. If $0 < \theta < 1$, $R(\theta)$ can still be decreased by moving θ to the right because it will be moving closer to three of the points—reducing their penalties—and away from only one of them. Similarly, if $2 < \theta$, $R(\theta)$ can be decreased by moving θ to the left. Only if $1 \leq \theta \leq 2$ is it impossible to decrease $R(\theta)$

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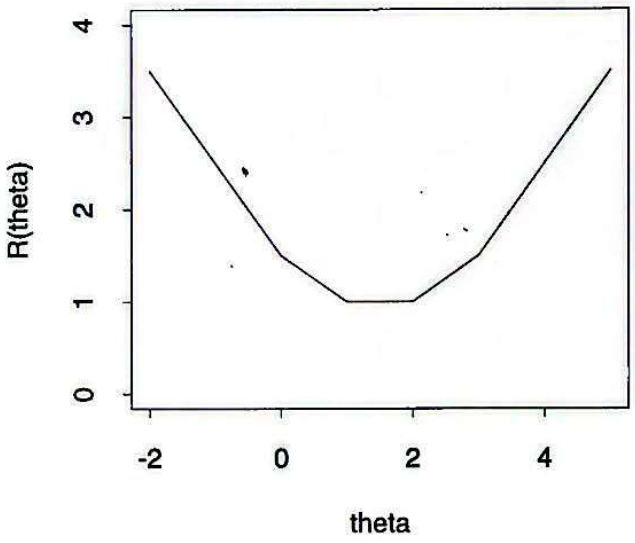


Figure 2.18: The risk using the absolute error loss function in Example 2.26.

by moving θ. But if $1 \leq \theta \leq 2$, there are as many possible values of x greater than or equal to θ as there are less than or equal to θ. Since all values of x_i are equally likely, θ satisfies (2.37).

Finally, consider situations in which there is no penalty for guessing x_i exactly (or within a small error ϵ) and a penalty of 1 for not getting the correct value. This leads to the **0–1 loss function**, which is defined by

$$L(\theta, x) = \begin{cases} 0 & \text{if } |x - \theta| < \epsilon \\ 1 & \text{otherwise.} \end{cases}$$

We choose θ to minimize

$$\begin{aligned} R(\theta) &= E[L(\theta, x)] = \sum_{|x_i - \theta| \geq \epsilon} P(x_i) \\ &= 1 - \sum_{|x_i - \theta| < \epsilon} P(x_i) \end{aligned}$$

If ϵ is small enough, the minimum of $R(\theta)$ will occur when $\theta = x_s$, where $P(x_s)$ is the largest of all the $P(x_i)$. The most likely value of x is called the **mode** (or dominant mode) of the distribution. Choosing θ equal to the most likely x_i maximizes its probability of being exactly correct (or within a small ϵ). The mode of a list is the most frequent number in it.

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Example 2.27 *Minimum risk estimates from a sample distribution.*

Even if the true distribution or density function of a random variable is unknown, the next outcome of the random variable x can be predicted by considering a sample of data with the values $x_1, x_2, \ldots, x_n$. Suppose the five samples 1, 1, 2, 5, and 6 are obtained from a random variable x. If the squared error loss function is used, the mean, 3, is the best guess for the next outcome of x. If the absolute error loss function is used, the best predicted value is the median, 2. Finally if the 0–1 loss function is used, the best guess is the mode 1.

Continuous Densities

If continuous densities are used, we obtain the same results as in the discrete case: The squared error loss function leads to the mean as the minimum risk predictor of the next outcome of the random variable; the absolute error loss function leads to the median; and the 0–1 loss function leads to the mode. The mean of a continuous random variable was defined in Section 3.5 as

$$\mu_x = E(x) = \int_{-\infty}^{\infty} xp(x)\,dx.$$

A **median** of a continuous random variable is any value m that divides the density of x into two equal areas, so that

$$P(x < m) = \int_{-\infty}^{m} p(x)\,dx = \int_{m}^{\infty} p(x)\,dx = P(x > m).$$

A **mode** is any value of x for which $p(x)$ attains a local maximum. If there is more than one mode, the x for which $p(x)$ is largest is called the **dominant mode**. A **local maximum** of a function is larger than any of the other values in a small interval centered at the local maximum. A local maximum is sometimes called a "peak." The dominant mode is the value of x at the highest peak.

The density function of a population consisting of a mixture of two normally distributed classes may have either one or two modes. For example, the graph of the density

$$p(x) = 0.75\left[\frac{1}{2\sqrt{2\pi}}e^{-(x-1)^2/8}\right] + 0.25\left[\frac{1}{\sqrt{2\pi}}e^{-(x-4)^2/2}\right]$$

is shown in Figure 2.19a. Its mean is $(0.75)(1) + (0.25)(4) = 1.75$ and its median can be found numerically to be 1.834. The density has two modes: The dominant mode is at $x = 3.52$ and the secondary mode is at $x = 1.12$. The median and the modes of

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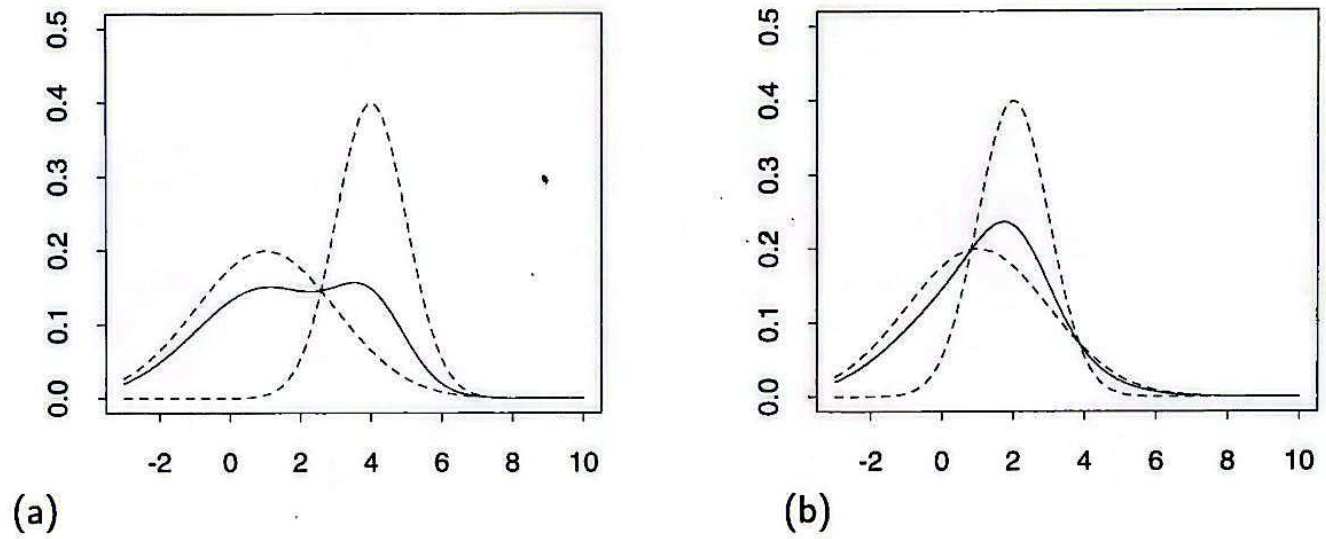


Figure 2.19: Mixture densities (solid curves) which are the weighted sum of two normal densities (dashed). $N(\mu, \sigma^2)$ denotes a normal density with mean μ and variance σ^2. (a) $p(x) = 0.75N(1,4) + 0.25N(4,1)$ is bimodal. (b) $p(x) = 0.8N(1,4) + 0.2N(2,1)$ is unimodal.

this type of function must usually be found by trial and error rather than analytically. Densities having two modes are called **bimodal densities**. The mixture density

$$p(x) = 0.8\left[\frac{1}{2\sqrt{2\pi}}e^{-(x-1)^2/8}\right] + 0.2\left[\frac{1}{\sqrt{2\pi}}e^{-(x-2)^2/2}\right]$$

is **unimodal** (has only one mode) as is shown in Figure 2.19b. Its mean, median, and mode are 1.2, 1.39, and 1.73, respectively.

Example 2.28 *Predicting the price of gold.*

Suppose that a commodities expert has reason to believe that the probability distribution for the price of gold one year from now, G, is given by the following table, and that the probability density is uniform within each of the five price ranges. Thus, price is a continuous random variable.

Event	P(Event)
$470 \leq G < 500$	0.1
$500 \leq G < 520$	0.2
$520 \leq G < 550$	0.3
$550 \leq G < 570$	0.3
$570 \leq G \leq 600$	0.1

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If he charges \$1,000 for his estimate, but returns \$20 to his clients for each dollar of error in his estimate, what price should he estimate in order to maximize his expected profit?

Since the penalty is proportional to the absolute error of the estimate, the median minimizes the expected loss. By definition, half of the area lies to the left of the median. The area to the left of 520 is 0.3 and the area between 520 and 550 is also 0.3, so the point that divides the area under the distribution in half is two-thirds of the way from 520 to 550. Thus, 540 is the best guess; that is, $0.1 + 0.2 + (2/3)(0.3) = 0.5$. If the density is correct, the guess of 540 would be too high half of the time and too low half of the time.

2.8 Problems

2.1. Suppose you want to set up a subjective probability model to predict the number of inches that it will rain tomorrow based on the weather forecast for tomorrow which says that "the probability of rain tomorrow is 30 percent." Assuming that you believe the forecast, fill in *your* subjective probabilities in the following table:

Amount in inches	Probability
$r = 0.0$	
$0.0 < r \leq 0.5$	
$0.5 < r \leq 1.0$	
$1.0 < r \leq 1.5$	
$1.5 < r \leq 2.0$	
$2.0 < r \leq 2.5$	
$2.5 < r$	

According to your table, what is the probability that it will rain more than one inch?

2.2. Two coins are flipped simultaneously. One has a probability of heads equal to 0.6 and the other has a probability of heads equal to 0.7. What is the probability that the coins will be both heads or both tails?

2.3. On a campus, 1/5 of the students are freshmen and 1/10 of the students are in engineering, but 1/4 of the engineering students are freshmen. What fraction of the students are neither engineers nor freshmen? Draw a Venn diagram.

2.4. For a very large group of students, the probability of receiving an A in the first English course was 30 percent and the probability of an A in the first calculus course was 20 percent. Also, 15 percent of the students had As in both classes.

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(a) Construct a table like the one in Example 2.3.

(b) What is the probability that a randomly selected student got exactly one A?

(c) What is the probability that a student got no As?

(d) If someone got an A in calculus, what is the probability that he or she got an A in English?

(e) Does the data precisely agree with the assumption that the event of getting an A in English is independent of the event of getting an A in calculus? [**Ans:** 0.2; 0.65; 0.75; no, $0.06 \neq 0.15$]

2.5. When does $P[(A \textit{ and } B)|C] = P(A|C)P(B|C)$?

2.6. In a sample of 10 parts, 7 passed an electrical test, 8 passed a mechanical test, and 6 passed both tests. Let E and M represent passing the electrical and mechanical tests.

(a) Construct a table like the one in Example 2.3.

(b) What are the values of $P(E \textit{ and } M)$, $P(E \textit{ and } \overline{M})$, $P(\overline{E} \textit{ and } M)$, and $P(\overline{E} \textit{ and } \overline{M})$ for the sample?

(c) What is the probability that a random part from the data set passed exactly one of the tests?

(d) What are the values of $P(E|M)$ and $P(E)$ for the sample? Are they exactly equal? Are E and M independent in this sample?

(e) Given this data, is it plausible that E and M are independent in the population?

2.7. Classes A and B are not mutually exclusive. In a data set of 11 samples, 7 samples belonged to class A, 5 belonged to class B, 2 belonged to both, and 1 belonged to neither. What is $P(A|B)$ for this data?

2.8. A pattern recognition system has been developed for classifying objects into six categories. It has been correct on 80 percent of the past testing data. What is the estimated probability that exactly four of the next five samples will be correctly classified?

2.9. A lab technician routinely misreads 10 percent of the tests made, on the average. Out of 12 tests, what is the probability that two or more are misread? *Hint*: Calculate the probability that none are misread and that exactly one is misread.

2.10. A classifier has a 30 percent error rate. What is the probability that exactly three errors will be made in classifying 10 samples? [**Ans:** 0.2668]

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2.11. A biased coin shows heads 60 percent of the time on the average. What is the probability that three of five flips will show heads?

2.12. A fair coin is tossed 50 times. What is the probability that heads will appear exactly 25 times?

2.13. A classifier has an average error rate of 3 errors per 50 samples. What is the probability that it will make exactly 3 errors on a set of 50 samples?

2.14. The **negative binomial distribution** gives the probability that exactly n trials will be required to attain r successes. For this distribution the number of successes is fixed, whereas with the binomial distribution, the number of trials is fixed and the probability of obtaining a certain number of successes is found. The distribution for the negative binomial is

$$P(n) = \binom{n-1}{k-1} \theta^k (1-\theta)^{n-k}$$

where θ is the probability of success on one trial.

A basketball player is an 80 percent successful free throw shooter. To make 20 free throws, what is the probability that he or she will need to shoot exactly 25 times?

2.15. Derive the distribution for the negative binomial. *Hint*: The last trial must be a success.

2.16. An X-ray source emits 10 photons per second on the average, with a Poisson distribution. What is the probability that exactly 10 photons will be emitted in a given second?

2.17. A large fire station responds on the average to four fires per day. If the distribution of the number of fires per day follows a Poisson distribution, what is the probability that there will be fewer than three fires tomorrow?

2.18. If $P(x)$ is the Poisson distribution, show that

$$\sum_{x=0}^{\infty} P(x) = 1.$$

2.19. If $p(x) = x/2$ for x ranging from 0 to 2, and $p(x) = 0$ otherwise, what is the mean of this density? Show your calculations and a graph of the density.

2.20. If $p(x) = 2x/9$ for $0 < x < 3$, and $p(x) = 0$ otherwise, what is $P(0 < x < 1)$? [**Ans:** 1/9]

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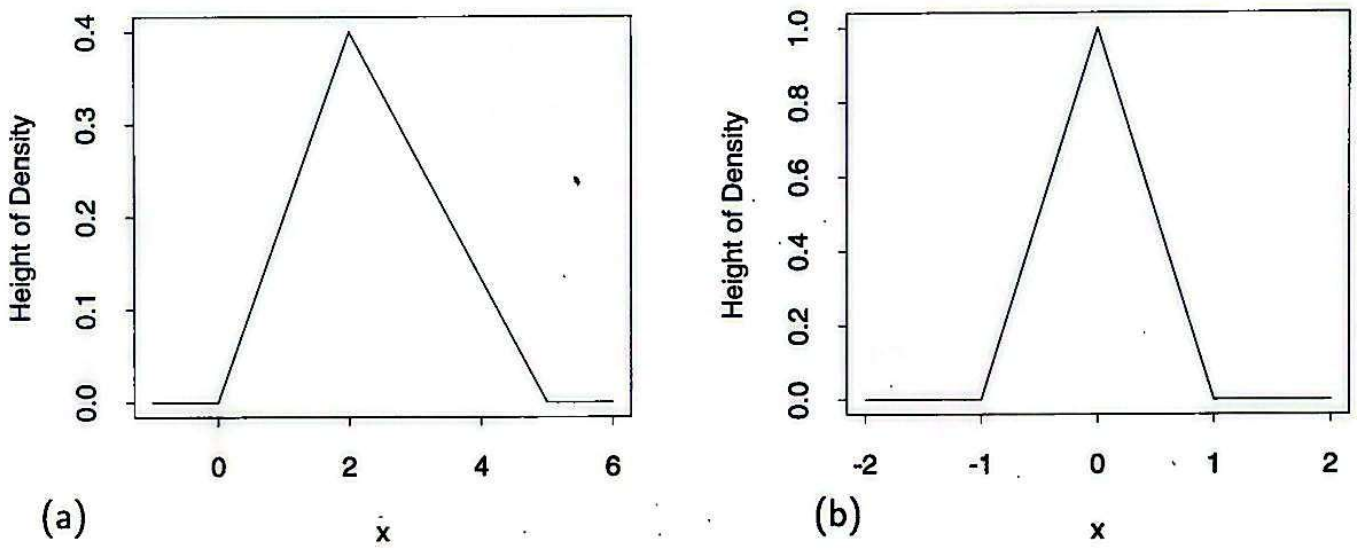


Figure 2.20: Triangular density functions for Problems 2.22 and 2.23.

2.21. Calculate $P(a \leq x \leq b)$ where x is a double exponential random variable with the distribution

$$p(x) = \frac{1}{2}e^{-|x|}.$$

Hint: Consider the two cases: a and b are both on the same side of 0, and a and b are on opposite sides of 0.

2.22. Feature x has an asymmetric triangular density with a peak at $x = 2$ and a range from 0 to 5 as shown in Figure 2.20a. Compute $P(1 < x < 2)$.

2.23. Calculate $P(-0.2 \leq x \leq 0.5)$ where x has the triangular density

$$p(x) = \begin{cases} 1 - x & \text{if } 0 \leq x \leq 1 \\ x + 1 & \text{if } -1 \leq x \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

shown in Figure 2.20b. [**Ans:** 0.555]

2.24. Calculate $P(a \leq x \leq b)$ where x has the Cauchy density

$$p(x) = \frac{1}{\pi(1 + x^2)}.$$

2.25. Feature x is normally distributed, with $\mu = 3$ and $\sigma = 2$. Find $P(-3 < x < -2)$.

2.26. Use the normal table in Figure 2.11 to calculate $P(0.5 \leq x \leq 1.5)$ if x has the standard normal density in (2.18).

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2.27. Use a z-transformation and the normal tables to calculate $P(-1 \leq x \leq 2)$ when x has the density

$$p(x) = \frac{1}{3\sqrt{2\pi}} e^{-(x-2)^2/18}.$$

[**Ans:** 0.3413]

2.28. Use the normal table in Figure 2.11 to find the probability that a normal random variable will be more than two standard deviations above its mean.

2.29. Show that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 1.$$

Hint: Note that

$$\left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz\right)^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-(x^2+y^2)/2}\, dy\, dx$$

and change to polar coordinates where $r = \sqrt{x^2+y^2}$.

2.30. For each of the following data sets, construct a normal plot, and decide if the data appear to be approximately normally distributed.

(a) 35, 43, 46, 48, 51, 55, 58, 65

(b) 2.0, 3.0, 3.2, 3.5, 3.7, 3.9, 4.0, 4.2, 4.4, 4.4, 4.5, 4.8, 5.0, 5.1, 5.4, 5.8, 6.1

2.31. Show that the multivariate normal density (2.21) becomes the bivariate normal density (2.29) when $d = 2$.

2.32. Show that if $\rho = 0$ in (2.21) then this density can be factored into the product of two one-dimensional normal densities.

2.33. Calculate the mean and variance of this data set: 2, 4, 4, 6. Calculate an unbiased estimate of the variance of the population from which this data was obtained. [**Ans:** 4, 2, 8/3]

2.34. Three samples have values of x of 1, 2, and 4 and are assumed to come from a normal distribution.

(a) What are the unbiased estimates of the mean and variance of x?

(b) What are the maximum likelihood estimates of the mean and variance of x?

2.35. Calculate the variance of the triangular density in Problem 2.20.

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2.36. Show that $\sigma_{x+y}^2 = \sigma_x^2 + 2\sigma_{xy} + \sigma_y^2$.

2.37. Find the variance of the density from Problem 2.20.

2.38. Write an expression for $\sigma_{x+y,z+w}$ in terms of σ_{xz}, σ_{yz}, σ_{xw}, and σ_{yw}.

2.39. Let x be a Poisson random variable with mean λ. Show that $\sigma_x^2 = \lambda$.

2.40. Three samples were drawn from a population that is believed to be uniformly distributed with an unknown range. The values of the feature x were 13, 20, and 27.

(a) What is the maximum likelihood estimate of the range of x?

(b) What is the method of moments estimate of the range of x?

(c) What is an unbiased estimate of the range of x?

2.41. Show that $\bar{x}$ is the maximum likelihood estimate of λ if x has the Poisson distribution with mean λ.

2.42. Let t be an exponential random variable with rate parameter β. Show that $\mathrm{E}(t) = 1/\beta$ and $\sigma_t^2 = 1/\beta^2$. *Hint*: Use integration by parts.

2.43. Show that $1/\bar{x}$ is the maximum likelihood estimate for θ if x has the exponential distribution with parameter θ.

2.44. Let x be a random variable with the double exponential density

$$p(x) = \frac{1}{2} e^{-|x-a|}.$$

Its mean is a because the density is symmetric about a. However, show that the maximum likelihood estimate of a is the median of the samples $x_1, \ldots, x_n$. *Hint*: The expression $\sum_{i=1}^n |x_i - a|$ is minimized when a is the median of $x_1, \ldots, x_n$.

2.45. Let x be a random variable with the double exponential density

$$p(x) = \frac{1}{2b} e^{-|x-a|/b},$$

where the parameter a (the mean) is known. Show that the maximum likelihood estimate for b is

$$\hat{b} = \frac{1}{n} \sum_{i=1}^{n} |x_i - a|.$$

The estimate $\hat{b}$ is called the **mean absolute deviation** (MAD).

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2.46. The joint probability density function of the samples $x_1, \ldots, x_n$ which come from a Cauchy distribution with $b = 1$ is

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} \frac{1}{1 + (x_i - a)^2}.$$

Plot $p(x_1, \ldots x_n)$ versus a for the set of samples $x_1 = 0$, $x_2 = 1$, and $x_3 = 4$. Find the value $\hat{a}$ that maximizes this function. Note that this maximum likelihood estimate of a is neither the mean nor the median of the samples, even though a is the median of the Cauchy density.

2.47. Show that the Cauchy distribution does not have a finite variance.

2.48. Find the maximum likelihood estimator of the range $b - a$ of a uniform density on the interval $[a, b]$, given two independent samples x_1 and x_2. *Hint*: Let $S = \min(x_1, x_2)$ and $L = \max(x_1, x_2)$. Show that the joint density of the independent samples x_1 and x_2 is

$$p(x_1, x_2) = p(x_1)p(x_2) = \begin{cases} 1/(b-a)^2 & \text{if } x_1, x_2 \in [a, b] \\ 0 & \text{otherwise.} \end{cases}$$

Now show that $p(x_1, x_2)$ is maximized when $a = S$ and $b = L$.

2.49. In Problem 2.48, show that an unbiased estimate of the width of the interval $b - a$ is $3(L - S)$ where $S = \min(x_1, x_2)$ and $L = \max(x_1, x_2)$. *Hint*: Verify that

$$\begin{aligned} \mathrm{E}(S) &= \int\int_{R^2} S p(x_1, x_2)\, dx_1\, dx_2 \\ &= \int\int_{x_1 \le x_2} x_1 p(x_1, x_2)\, dx_1 dx_2 + \int\int_{x_2 \le x_1} x_2 p(x_1, x_2)\, dx_2 dx_1 \\ &= \int_a^b \int_a^{x_2} x_1 \frac{dx_1\, dx_2}{(b-a)^2} + \int_a^b \int_a^{x_1} x_2 \frac{dx_2\, dx_1}{(b-a)^2} = a + (1/3)(b-a). \end{aligned}$$

Similarly, show that $\mathrm{E}(L) = a + (2/3)(b-a)$ and then find $\mathrm{E}(L-S) = \mathrm{E}(L) - \mathrm{E}(S)$.

2.50. Let $x_1, \ldots, x_n$ be samples from a density function with parameters μ and σ^2. If the samples are chosen independently, the expected covariance between each pair of random variables is zero; that is, $E[(x_i - \mu)(x_j - \mu)] = 0$. Show that if

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2,$$

then $\mathrm{E}(s^2) = \sigma^2$. Use the following steps:

(a) From equation (2.24), $\mathrm{E}(x_i^2) = \mu^2 + \sigma^2$.

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(b) Show that $\mathrm{E}(x_i x_j) = \mu^2$.

(c) Show that $\mathrm{E}(x_i \bar{x}) = \mu^2 + (1/n)\sigma^2$.

(d) Show that $\mathrm{E}(\bar{x}^2) = \mu^2 + (1/n)\sigma^2$.

(e) Calculate $\mathrm{E}\left[\sum_{i=1}^{n}(x_i - \bar{x})^2\right]$.

2.51. Show that if the random variable x with density $p(x)$ is symmetric around the origin ($p(x) = p(-x)$) and $\mathrm{E}(x)$ is finite, then $\mathrm{E}(x) = 0$.

2.52. Let the random variable x have the density $p(x)$ defined as $p(x) = 2x/9$ when $0 < x < 3$, and $p(x) = 0$ elsewhere.

(a) Compute the third central moment. [**Ans:** -0.2]

(b) Compute the fourth central moment.

2.53. Assume that three samples of the feature vector (x, y) are $(2, 9)$, $(5, 3)$, and $(3, 7)$. Make a scatterplot of this data in the xy-plane. Estimate the parameters μ_x, μ_y, σ_x^2, σ_y^2, and σ_{xy} using the unbiased estimates. Estimate ρ_{xy} using the values you obtained for σ_x^2, σ_y^2, and σ_{xy}. [**Ans:** 10/3, 19/3, 7/3, 28/3, $-14/3$, -1]

2.54. Solve Problem 2.53 given the sample feature vectors $(1, 3)$, $(2, 5)$, and $(3, 6)$.

2.55. If x is normally distributed with mean 2 and variance 4, and y is normally distributed with mean 0 and variance 1, and x and y are independent, what is

$$P(2 \leq x < 3 \text{ and } 0 \leq y < 1)?$$

2.56. Suppose that the values 2, 3, 4, 5, 6, and 16 came from a uniform distribution. Use the method of moments (see Example 2.19) to estimate the values a and b that determine the range $[a, b]$. Notice that at least one of the samples lies outside the estimated range in this case.

2.57. Consider the mixture density $0.5N(-\mu, 1) + 0.5N(\mu, 1)$. What is the largest value of μ for which the density is unimodal? *Hint*: The second derivative of the density is positive at $x = 0$ when the density is bimodal and negative when it is unimodal.

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